Model Question Paper – Cloud Data Engineer- Jain University

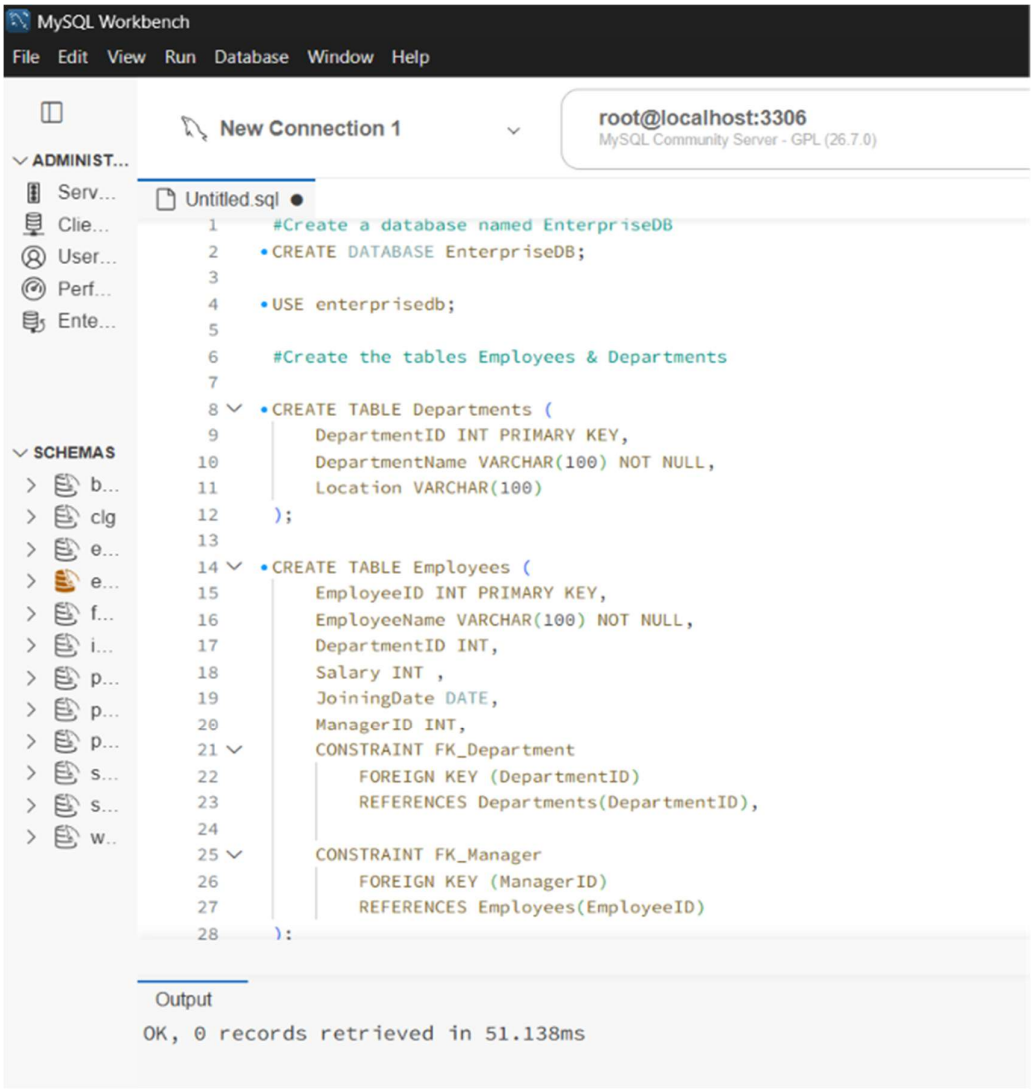
HCL ID: HCLTFP2170249

NAME: DHARUN P

USN: 23BTRCL014

DEPARTMENT: AIML

SECTION A: ADVANCED SQL || Maximum Marks: 20 Question 1: Database Design and Advanced Queries (8 Marks)



New Connection 1

root@localhost:3306
MySQL Community Server - GPL (26.7.0)

Untitled.sql

```
29
30    #Insert at least 15 employee records and five department records
31
32   •INSERT INTO Departments VALUES
33    (1,'HR','Bangalore'),
34    (2,'IT','Chennai'),
35    (3,'Finance','Mumbai'),
36    (4,'Sales','Delhi'),
37    (5,'Marketing','Hyderabad');
38
39
40   •INSERT INTO Employees VALUES
41    (101,'Ram',2,90000,'2023-01-10',NULL),
42    (102,'Ragul',1,70000,'2023-02-01',101),
43    (103,'Amit',2,65000,'2024-03-05',101),
44    (104,'Neha',3,85000,'2022-06-10',101),
45    (105,'Ravi',4,55000,'2024-01-12',102),
46    (106,'Pooja',4,60000,'2023-10-20',102),
47    (107,'Kiran',3,95000,'2021-05-15',104),
48    (108,'Anita',2,72000,'2024-02-17',101),
49    (109,'Vijay',1,50000,'2024-03-01',102),
50    (110,'Deepa',3,88000,'2023-12-15',104),
51    (111,'Suresh',4,47000,'2024-01-01',105),
52    (112,'Nisha',2,79000,'2023-07-08',101),
53    (113,'Arjun',3,99000,'2022-02-02',104),
54    (114,'Meera',1,76000,'2024-05-15',102),
55    (115,'Akash',2,68000,'2024-06-12',101);
56
```

Tasks: a. Display employee names, department names, and manager names using joins. (2 marks)

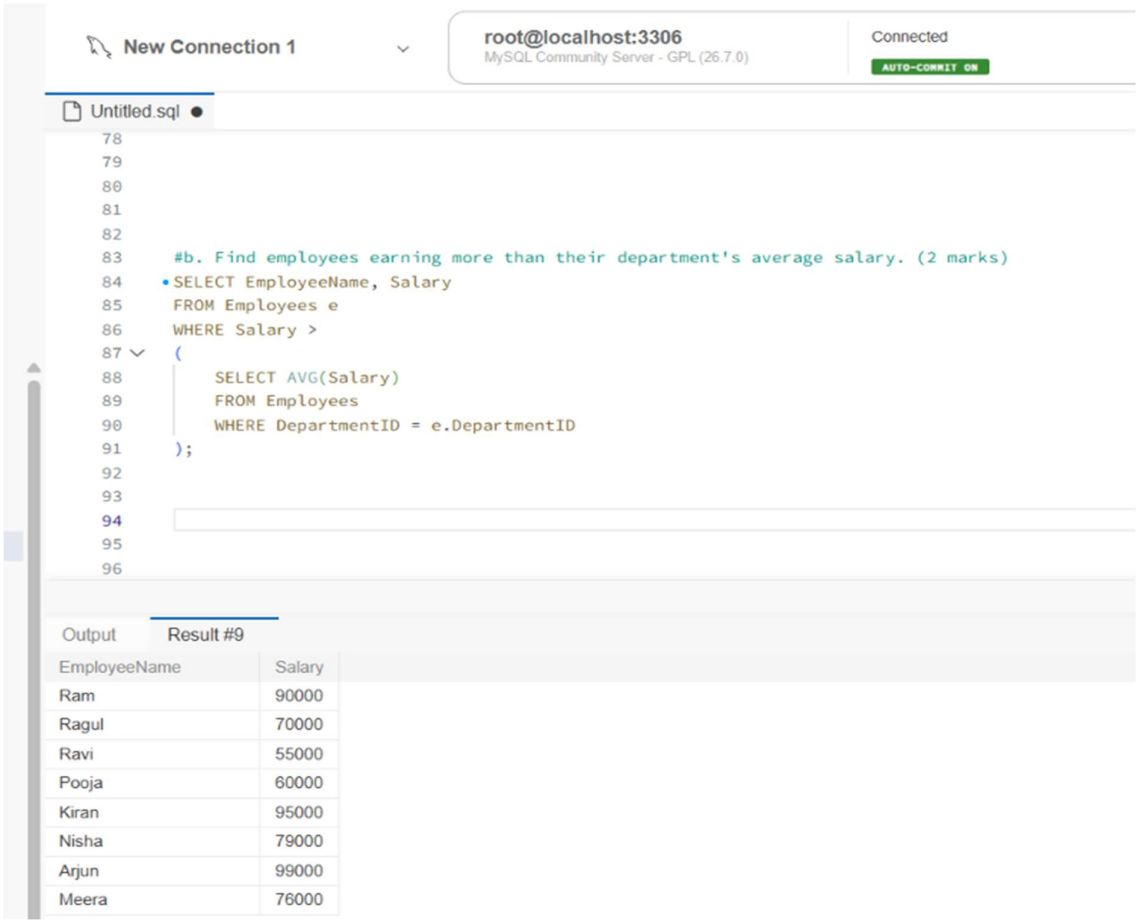
Untitled.sql

```
66
67    #Tasks:
68    #a. Display employee names, department names, and manager names using joins. (2marks)
69   •SELECT e.EmployeeName,
70           d.DepartmentName,
71           m.EmployeeName AS ManagerName
72    FROM Employees e
73    LEFT JOIN Departments d
74    ON e.DepartmentID = d.DepartmentID
75    LEFT JOIN Employees m
76    ON e.ManagerID = m.EmployeeID;
77
78
79
80
```

Output | Result #8

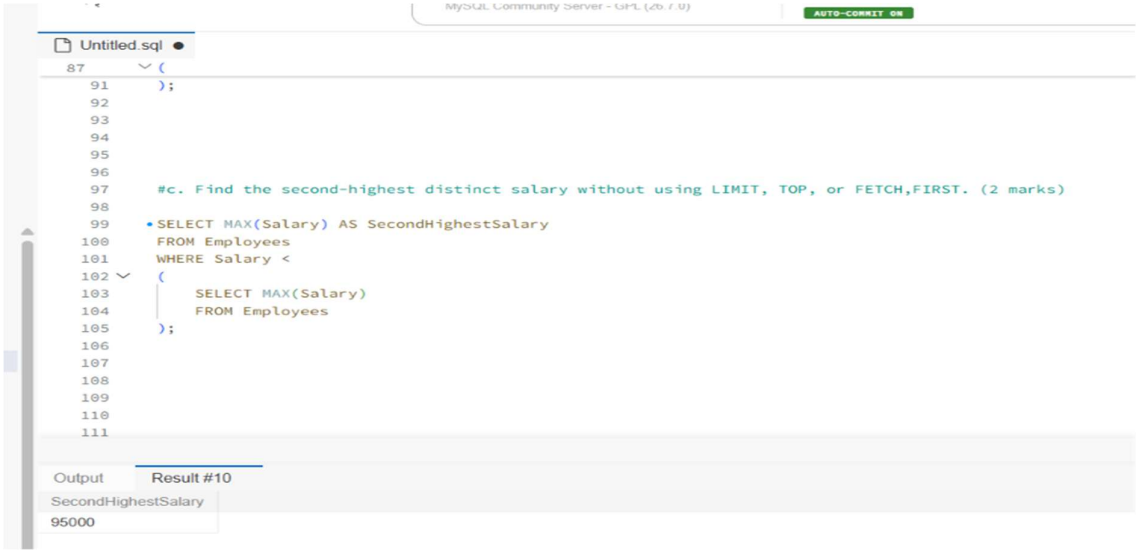
EmployeeName	DepartmentName	ManagerName
Ram	IT	NULL
Ragul	HR	Ram
Amit	IT	Ram
Neha	Finance	Ram
Ravi	Sales	Ragul
Pooja	Sales	Ragul
Kiran	Finance	Neha
Anita	IT	Ram
Vijay	HR	Ragul
Deepa	Finance	Neha
Suresh	Sales	Ravi
Nisha	IT	Ram
Arjun	Finance	Neha
Meera	HR	Ragul
Akash	IT	Ram

b. Find employees earning more than their department's average salary. (2 marks)



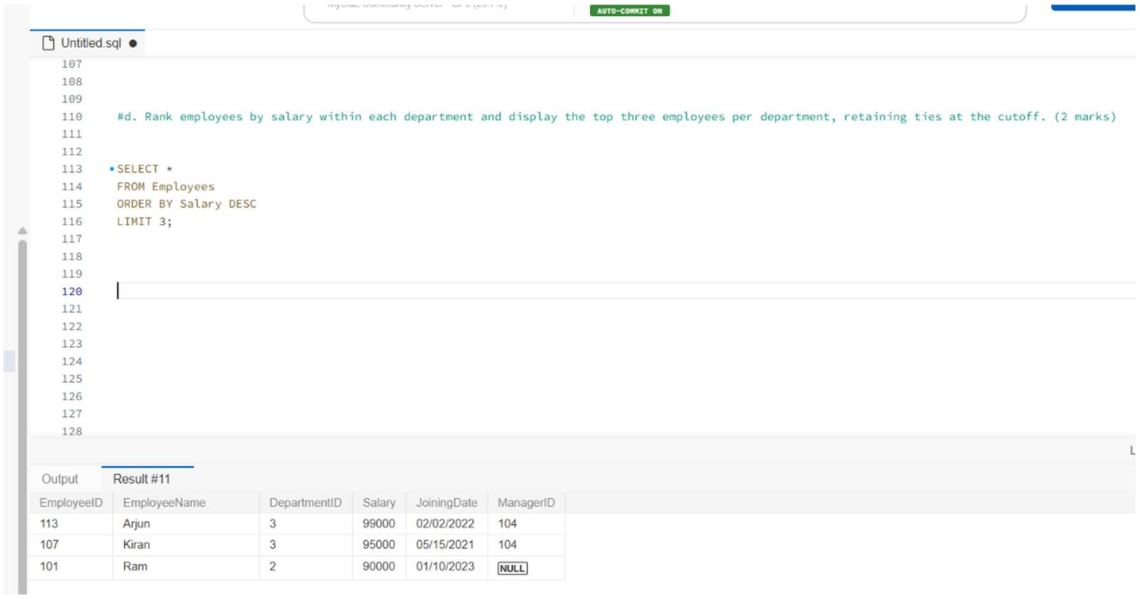
EmployeeName	Salary
Ram	90000
Ragul	70000
Ravi	55000
Pooja	60000
Kiran	95000
Nisha	79000
Arjun	99000
Meera	76000

c. Find the second-highest distinct salary without using LIMIT, TOP, or FETCH FIRST. (2 marks)



SecondHighestSalary
95000

d. Rank employees by salary within each department and display the top three employees per department, retaining ties at the cutoff. (2 marks)



Untitled.sql

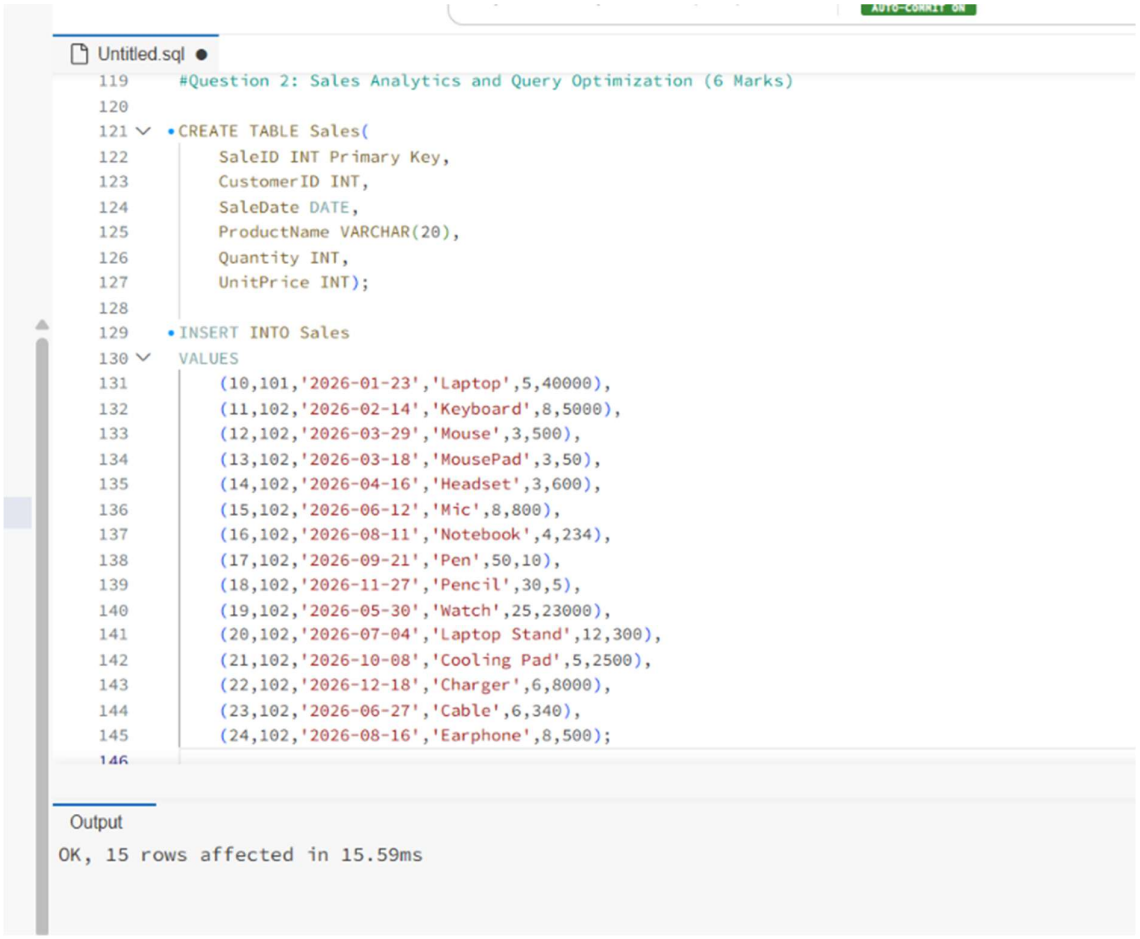
```
#d. Rank employees by salary within each department and display the top three employees per department, retaining ties at the cutoff. (2 marks)

SELECT *
FROM Employees
ORDER BY Salary DESC
LIMIT 3;
```

Output | Result #11

EmployeeID	EmployeeName	DepartmentID	Salary	JoiningDate	ManagerID
113	Arjun	3	99000	02/02/2022	104
107	Kiran	3	95000	05/15/2021	104
101	Ram	2	90000	01/10/2023	NULL

Question 2: Sales Analytics and Query Optimization (6 Marks)



Untitled.sql

```
#Question 2: Sales Analytics and Query Optimization (6 Marks)

CREATE TABLE Sales(
    SaleID INT Primary Key,
    CustomerID INT,
    SaleDate DATE,
    ProductName VARCHAR(20),
    Quantity INT,
    UnitPrice INT);

INSERT INTO Sales
VALUES
    (10,101,'2026-01-23','Laptop',5,40000),
    (11,102,'2026-02-14','Keyboard',8,5000),
    (12,102,'2026-03-29','Mouse',3,500),
    (13,102,'2026-03-18','MousePad',3,50),
    (14,102,'2026-04-16','Headset',3,600),
    (15,102,'2026-06-12','Mic',8,800),
    (16,102,'2026-08-11','Notebook',4,234),
    (17,102,'2026-09-21','Pen',50,10),
    (18,102,'2026-11-27','Pencil',30,5),
    (19,102,'2026-05-30','Watch',25,23000),
    (20,102,'2026-07-04','Laptop Stand',12,300),
    (21,102,'2026-10-08','Cooling Pad',5,2500),
    (22,102,'2026-12-18','Charger',6,8000),
    (23,102,'2026-06-27','Cable',6,340),
    (24,102,'2026-08-16','Earphone',8,500);
```

Output

```
OK, 15 rows affected in 15.59ms
```

a. Calculate total revenue for each product. (2 marks)

```
147
148  #a. Calculate total revenue for each product. (2 marks)
149
150
151  SELECT ProductName,Quantity,UnitPrice,(Quantity * UnitPrice) AS total_revenue
152  FROM Sales;
153
154
155
156
```

Output | Result #14

ProductName	Quantity	UnitPrice	total_revenue
Laptop	5	40000	200000
Keyboard	8	5000	40000
Mouse	3	500	1500
MousePad	3	50	150
Headset	3	600	1800
Mic	8	800	6400
Notebook	4	234	936
Pen	50	10	500
Pencil	30	5	150
Watch	25	23000	575000
Laptop Stand	12	300	3600
Cooling Pad	5	2500	12500
Charger	6	8000	48000
Cable	6	340	2040
Earphone	8	500	4000

b. Calculate monthly revenue and month-over-month revenue change using a window function. (2 marks)

```
156  #b. Calculate monthly revenue and month-over-month revenue change using a window function. (2 marks
157
158  WITH MonthlyRevenue AS
159  (
160      SELECT
161          YEAR(SaleDate) AS SalesYear,
162          MONTH(SaleDate) AS SalesMonth,
163          SUM(Quantity * UnitPrice) AS Revenue
164      FROM Sales
165      GROUP BY YEAR(SaleDate), MONTH(SaleDate)
166  )
167  SELECT *,
168          Revenue -
169          LAG(Revenue) OVER
170          (ORDER BY SalesYear, SalesMonth)
171          AS RevenueChange
172  FROM MonthlyRevenue;
```

Output | Result #15

SalesYear	SalesMonth	Revenue	RevenueChange
2026	1	200000	NULL
2026	2	40000	-160000
2026	3	1650	-38350
2026	4	1800	150
2026	5	575000	573200
2026	6	8440	-566560
2026	7	3600	-4840
2026	8	4936	1336
2026	9	500	-4436
2026	10	12500	12000
2026	11	150	-12350
2026	12	48000	47850

c. **Rewrite the following query to improve the potential for index usage and explain why your version may perform better. (2 marks)**
SELECT * FROM Employees
WHERE YEAR(JoiningDate) = 2025;
Assume JoiningDate may be indexed.

```
#c. Rewrite the following query to improve the potential for index usage and explain why your version may perform better. (2 marks)
#SELECT *
#FROM Employees
#WHERE YEAR(JoiningDate) = 2025;
#Assume JoiningDate may be indexed.

SELECT *
FROM Employees
WHERE JoiningDate >= '2023-01-01'
AND JoiningDate < '2024-01-01';
```

Output | Result #16

EmployeeID	EmployeeName	DepartmentID	Salary	JoiningDate	ManagerID
101	Ram	2	90000	01/10/2023	NULL
102	Ragul	1	70000	02/01/2023	101
106	Pooja	4	60000	10/20/2023	102
110	Deepa	3	88000	12/15/2023	104
112	Nisha	2	79000	07/08/2023	101

Question 3: SQL Troubleshooting (6 Marks)

A company maintains millions of employee and transaction records. Its reporting queries are running slowly. Perform the following tasks:

a. **Explain how an execution plan helps identify expensive operations in a query. (2 marks)**

An execution plan shows how the database executes a SQL query.

It helps us:

- Understand query performance
- Identify slow operations
- Check whether indexes are being used
- Find bottlenecks in a query

b. Recommend an appropriate index strategy for queries filtering employees by DepartmentID and Salary. Explain your reasoning. (2 marks)

```
SELECT *
FROM Employees
WHERE DepartmentID = 2;
```

Create an index on DepartmentID.

```
CREATE INDEX idx_department

ON Employees(DepartmentID);
```

c. Write a query to identify duplicate customer email addresses and display only addresses occurring more than once. (2 marks)

```
SELECT EmailAddress,
COUNT(*)
FROM Customers
GROUP BY EmailAddress
HAVING COUNT(*) > 1;
```

SECTION B: LINUX || Maximum Marks: 15

Question 4: Log Analysis (5 Marks) Assume /var/log/application.log contains millions of log entries in the following format:
2026-10-28 10:15:23 ERROR payment-service Transaction failed
Write Linux commands to:

a.Find all lines containing ERROR. (1 mark)

ANSWER: grep "ERROR" application.log

b. Display the 20 most recent error entries in the file. (1 mark)

ANSWER: grep "ERROR" application.log | tail -20

c. Count the total number of error entries. (1 mark)

ANSWER: grep -c "ERROR" application.log

d. Extract error entries for a specified date into a separate file. (1 mark)

ANSWER: grep "2025-01-10" application.log > output.log

e. Explain how you would calculate the number of errors occurring in each hour. (1 mark)

ANSWER: grep "ERROR" application.log | cut -d':' -f1 | sort | uniq -c

Question 5: Permissions and System Troubleshooting (5 Marks) A Linux server hosts an application that processes confidential financial reports.

Perform the following tasks:

a. Create a group named finance and a report file, assuming you have the required privileges. (1 mark)

sudo groupadd finance

touch report.txt

b. Set permissions so the owner can read and write, the finance group can read, and other users have no access. (1 mark)

chgrp finance report.txt

chmod 640 report.txt

c.Identify the processes consuming the most CPU. (1 mark)

ps aux --sort=-%cpu | head -6

d.Display memory usage and filesystem disk utilization. (1 mark)

memory: free -h

disk: df -h

e. Find files larger than 100 MB under a specified application directory. (1 mark)

find /app -type f -size +100M

SECTION C: PYSPARK DATA ENGINEERING || Maximum Marks: 20

Question 7: PySpark DataFrame Operations (6 Marks)

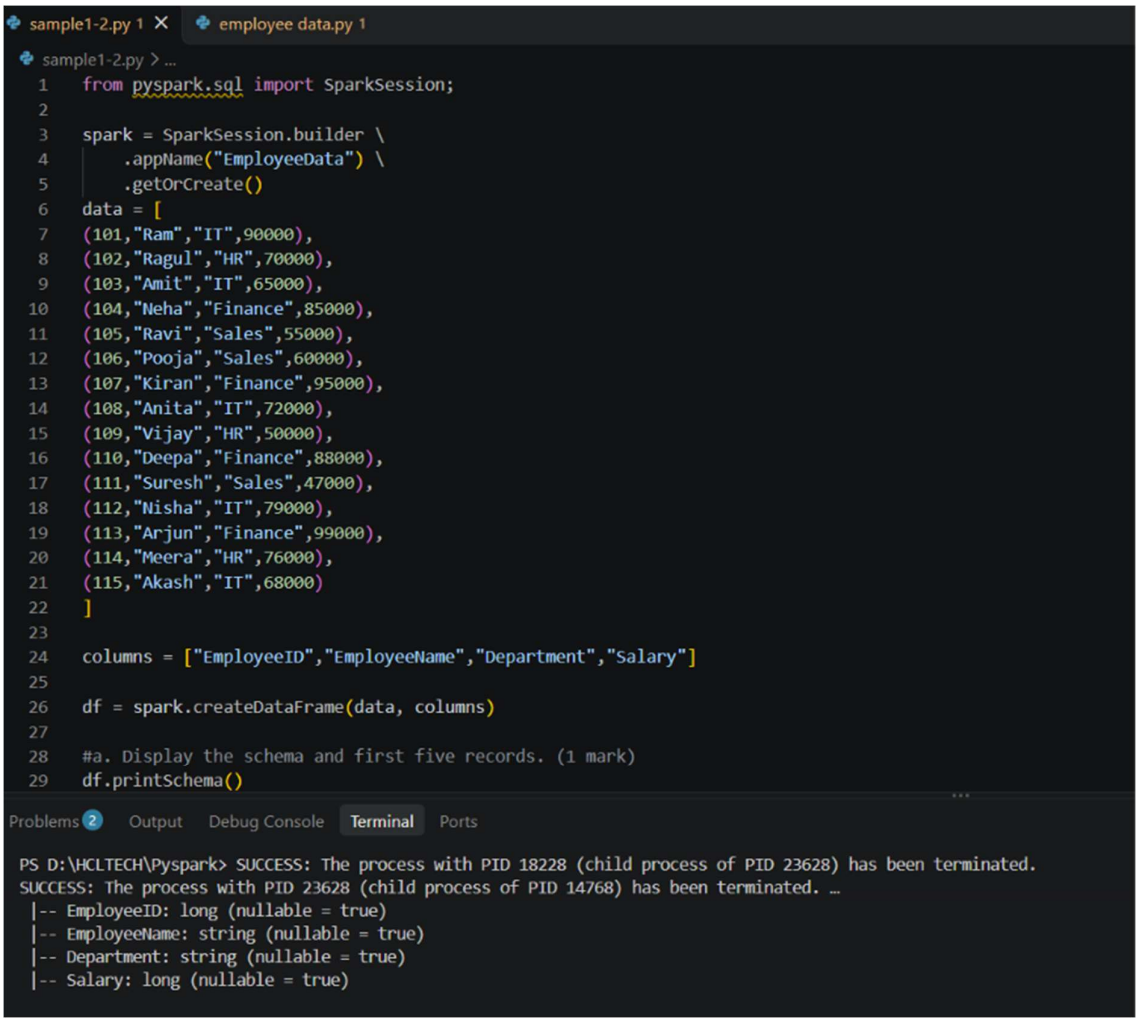
Create a PySpark DataFrame containing the following columns:

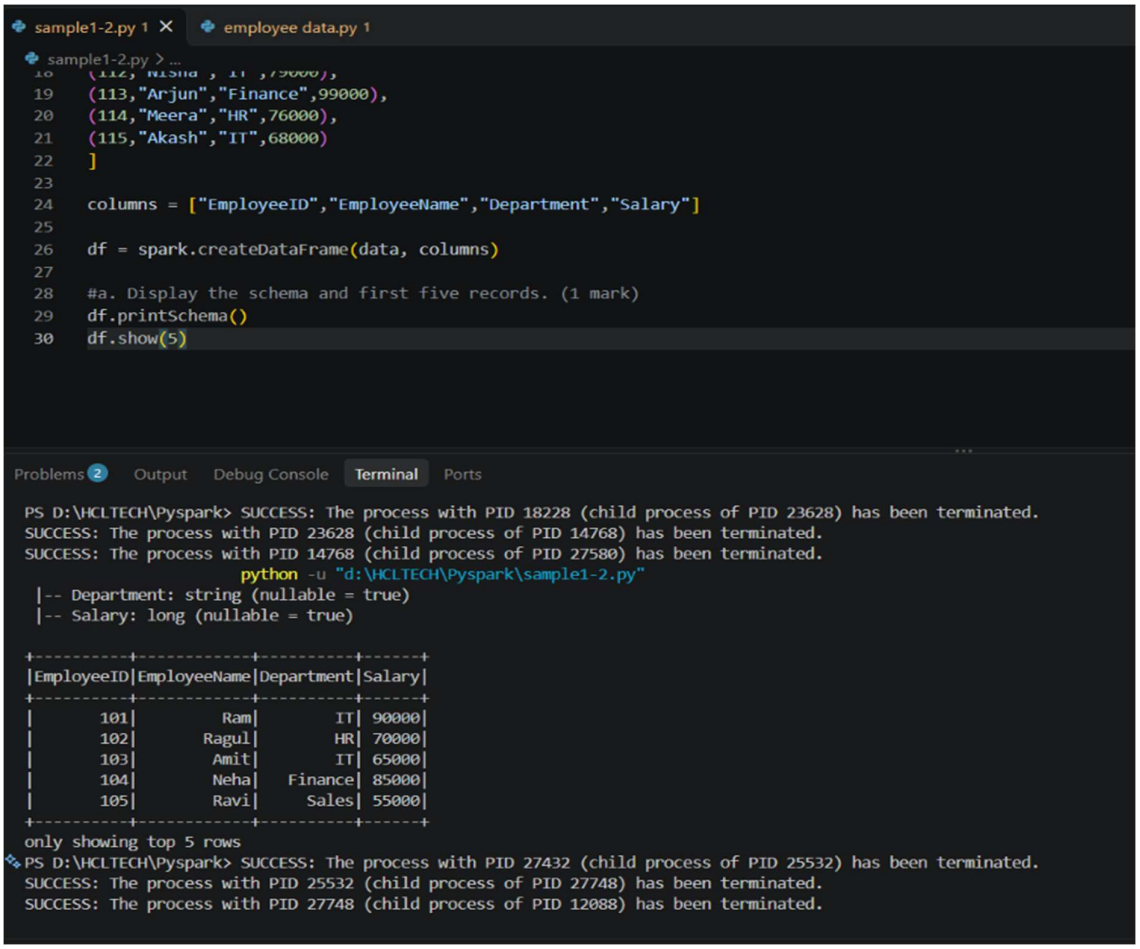
- **EmployeeID**
- **EmployeeName**

- **Department**
- **Salary**
- **JoiningDate Insert at least 15 sample records.**

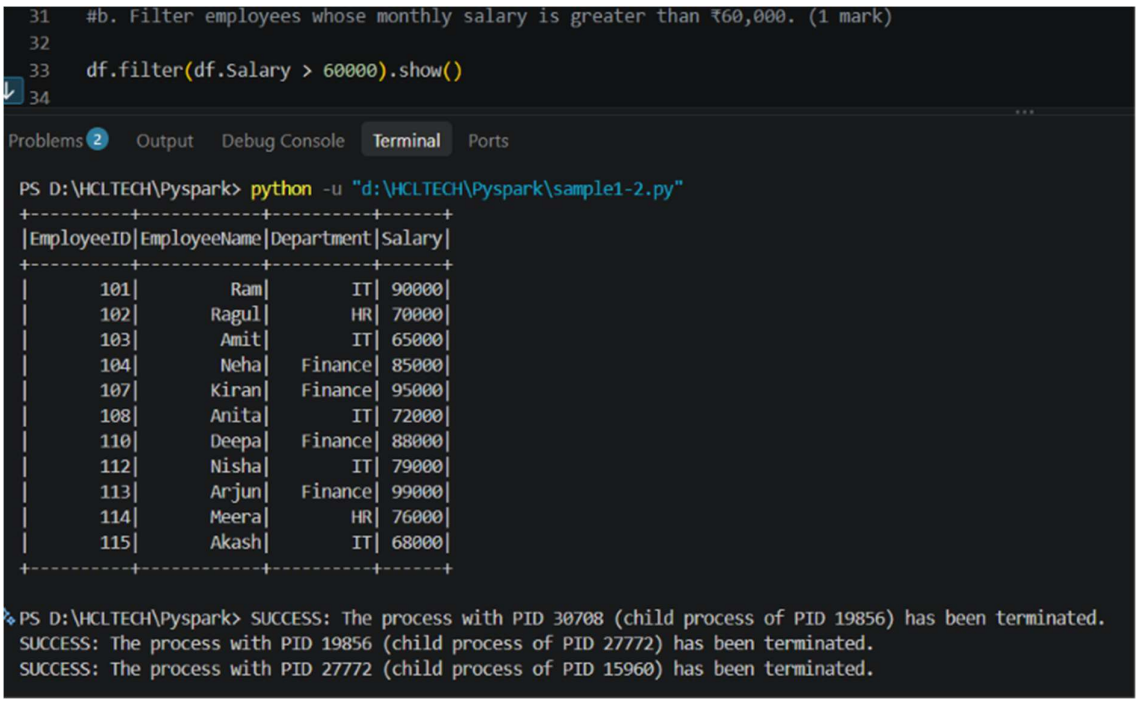
Write and execute a PySpark program to:

a. Display the schema and first five records. (1 mark)





b. Filter employees whose monthly salary is greater than ₹60,000. (1 mark)



c. Add an AnnualSalary column calculated from the monthly salary. (1 mark)

```
38   #c. Add an AnnualSalary column calculated from the monthly salary. (1 mark)
39   df = df.withColumn("AnnualSalary", df.Salary * 12)
40   df.show()
```

Problems 2 | Output | Debug Console | Terminal | Ports

```
PS D:\HCLTECH\Pyspark> SUCCESS: The process with PID 30708 (child process of PID 19856) has been terminated.
SUCCESS: The process with PID 19856 (child process of PID 27772) has been terminated.
SUCCESS: The process with PID 27772 (child process of PID 15960) has been terminated.
                    python -u "d:\HCLTECH\Pyspark\sample1-2.py"

+----------+------------+----------+------+------------+
|EmployeeID|EmployeeName|Department|Salary|AnnualSalary|
+----------+------------+----------+------+------------+
|       101|         Ram|        IT| 90000|     1080000|
|       102|       Ragul|        HR| 70000|      840000|
|       103|        Amit|        IT| 65000|      780000|
|       104|        Neha|   Finance| 85000|     1020000|
|       105|        Ravi|     Sales| 55000|      660000|
|       106|       Pooja|     Sales| 60000|      720000|
|       107|       Kiran|   Finance| 95000|     1140000|
|       108|       Anita|        IT| 72000|      864000|
|       109|       Vijay|        HR| 50000|      600000|
|       110|       Deepa|   Finance| 88000|     1056000|
|       111|      Suresh|     Sales| 47000|      564000|
|       112|       Nisha|        IT| 79000|      948000|
|       113|       Arjun|   Finance| 99000|     1188000|
|       114|       Meera|        HR| 76000|      912000|
|       115|       Akash|        IT| 68000|      816000|
+----------+------------+----------+------+------------+
```

d. Group employees by department and calculate the average salary and employee count. (2 marks)

```
45   #d. Group employees by department and calculate the average salary and employee count. (2 marks)
46
47   df.groupBy("Department").agg({"Salary": "avg", "EmployeeID": "count"}).show()
48
```

Problems 2 | Output | Debug Console | Terminal | Ports

```
PS D:\HCLTECH\Pyspark> python -u "d:\HCLTECH\Pyspark\sample1-2.py"

+----------+------------------+-----------------+
|Department|       avg(Salary)|count(EmployeeID)|
+----------+------------------+-----------------+
|        IT|           74800.0|                5|
|        HR|65333.333333333336|                3|
|   Finance|           91750.0|                4|
|     Sales|           54000.0|                3|
+----------+------------------+-----------------+
```

e. Display the five highest-paid employees. (1 mark)

```
52   #e. Display the five highest-paid employees. (1 mark)
53   df.orderBy(df.Salary.desc()).show(5)
```

Problems 2 | Output | Debug Console | Terminal | Ports

```
PS D:\HCLTECH\Pyspark> python -u "d:\HCLTECH\Pyspark\sample1-2.py"

+----------+------------+----------+------+------------+
|EmployeeID|EmployeeName|Department|Salary|AnnualSalary|
+----------+------------+----------+------+------------+
|       113|       Arjun|   Finance| 99000|     1188000|
|       107|       Kiran|   Finance| 95000|     1140000|
|       101|         Ram|        IT| 90000|     1080000|
|       110|       Deepa|   Finance| 88000|     1056000|
|       104|        Neha|   Finance| 85000|     1020000|
+----------+------------+----------+------+------------+
only showing top 5 rows
PS D:\HCLTECH\Pyspark> pSUCCESS: The process with PID 24704 (child process of PID 26492) has been terminated.
SUCCESS: The process with PID 26492 (child process of PID 16436) has been terminated.
SUCCESS: The process with PID 16436 (child process of PID 20120) has been terminated.
```

Question 8: Data Cleaning and Aggregation (7 Marks)

A company receives a CSV file named transactions.csv containing:

- **TransactionID**
- **CustomerID**
- **TransactionDate**
- **Product**
- **Quantity**
- **UnitPrice The file contains duplicate records, missing values, and incorrectly formatted dates. Write a PySpark program to:**

question8.py > ...

```python
from pyspark.sql import SparkSession
from pyspark.sql.functions import to_date, sum

# Create Spark Session
spark = SparkSession.builder \
    .appName("Transaction Analysis") \
    .getOrCreate()

# Read CSV File
df = spark.read.csv(
    "transactions.csv",
    header=True,
    inferSchema=True
)

print("Original Data")
df.show()

# Remove Duplicate Records
df = df.dropDuplicates()

print("After Removing Duplicates")
df.show()

# Remove Rows with Null Values
df = df.na.drop()

print("After Removing Null Values")
df.show()

# Convert TransactionDate to Date Format
df = df.withColumn(
    "TransactionDate",
    to_date("TransactionDate", "yyyy-MM-dd")
)

print("After Date Conversion")
df.show()
```

a. Read the CSV file using an appropriate schema and header configuration. (1 mark)

b. Identify and remove duplicate transactions using a suitable business key. (1 mark)

c. Handle missing values in required fields and explain your chosen approach. (2 marks)

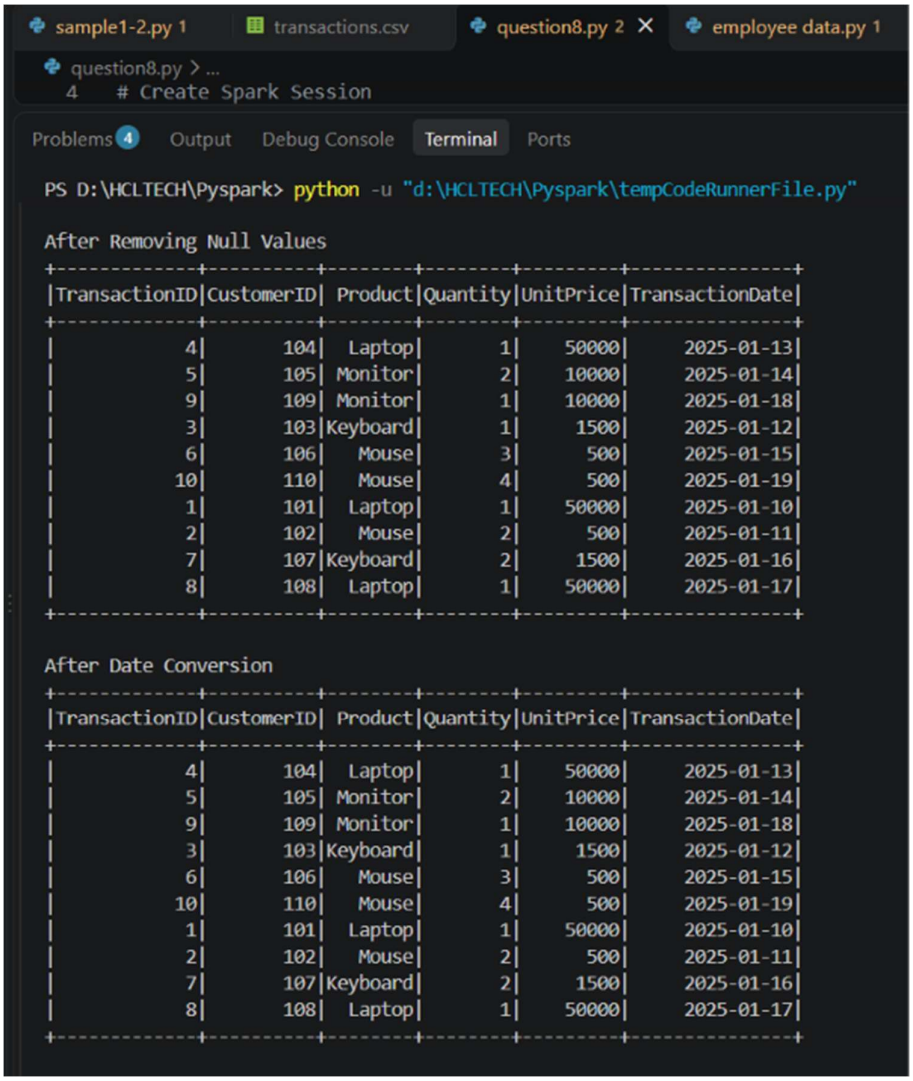
d. Convert TransactionDate into a date type and handle invalid dates. (1 mark)

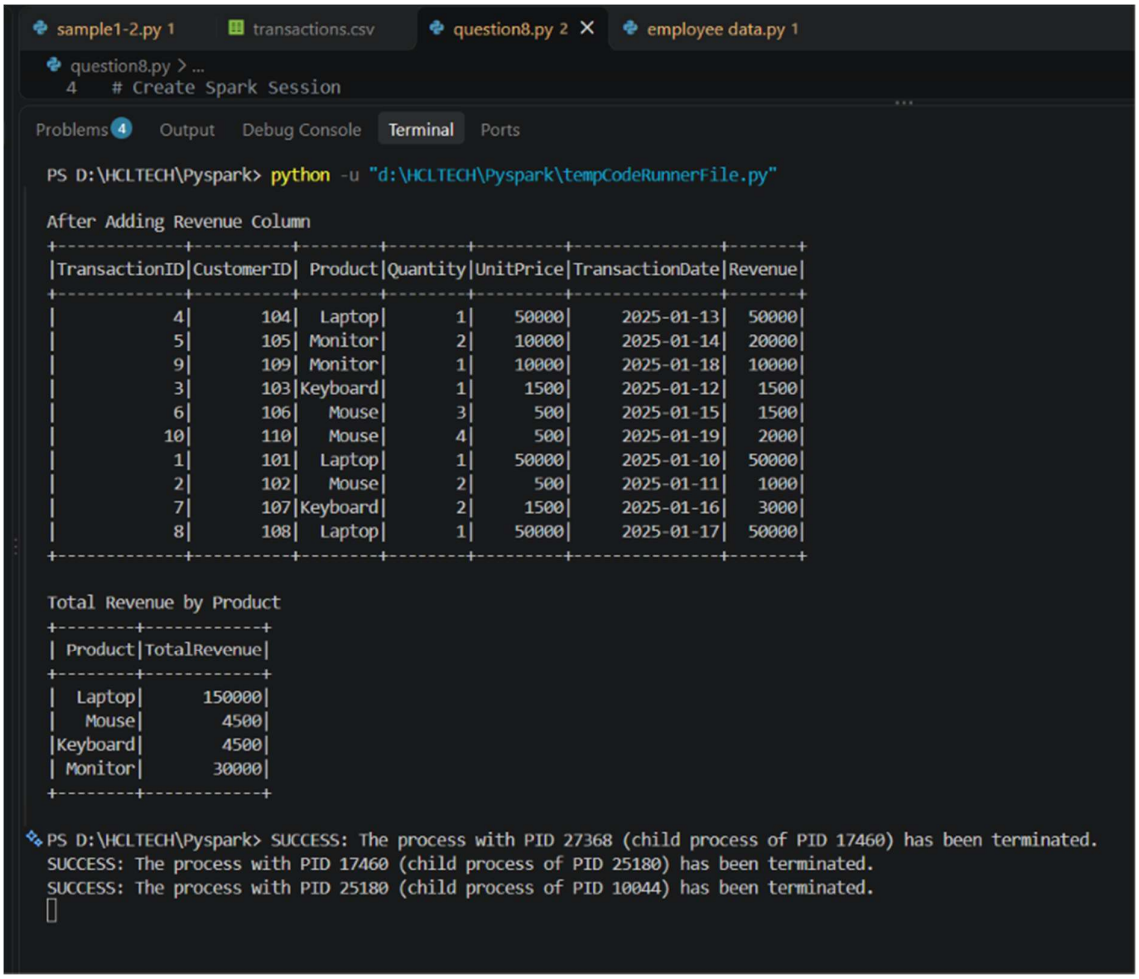
e. Calculate revenue using Quantity × UnitPrice. (1 mark)

f. Calculate total revenue by product and display the results in descending order. (1 mark)

```
PS D:\HCLTECH\Pyspark> python -u "d:\HCLTECH\Pyspark\tempCodeRunnerFile.py"
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/10/28 12:46:08 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Original Data
+-------------+----------+--------+--------+---------+---------------+
|TransactionID|CustomerID| Product|Quantity|UnitPrice|TransactionDate|
+-------------+----------+--------+--------+---------+---------------+
|            1|       101|  Laptop|       1|    50000|     2025-01-10|
|            2|       102|   Mouse|       2|      500|     2025-01-11|
|            3|       103|Keyboard|       1|     1500|     2025-01-12|
|            4|       104|  Laptop|       1|    50000|     2025-01-13|
|            5|       105| Monitor|       2|    10000|     2025-01-14|
|            6|       106|   Mouse|       3|      500|     2025-01-15|
|            7|       107|Keyboard|       2|     1500|     2025-01-16|
|            8|       108|  Laptop|       1|    50000|     2025-01-17|
|            9|       109| Monitor|       1|    10000|     2025-01-18|
|           10|       110|   Mouse|       4|      500|     2025-01-19|
+-------------+----------+--------+--------+---------+---------------+

After Removing Duplicates
+-------------+----------+--------+--------+---------+---------------+
|TransactionID|CustomerID| Product|Quantity|UnitPrice|TransactionDate|
+-------------+----------+--------+--------+---------+---------------+
|            4|       104|  Laptop|       1|    50000|     2025-01-13|
|            5|       105| Monitor|       2|    10000|     2025-01-14|
|            9|       109| Monitor|       1|    10000|     2025-01-18|
|            3|       103|Keyboard|       1|     1500|     2025-01-12|
|            6|       106|   Mouse|       3|      500|     2025-01-15|
|           10|       110|   Mouse|       4|      500|     2025-01-19|
|            1|       101|  Laptop|       1|    50000|     2025-01-10|
|            2|       102|   Mouse|       2|      500|     2025-01-11|
|            7|       107|Keyboard|       2|     1500|     2025-01-16|
|            8|       108|  Laptop|       1|    50000|     2025-01-17|
+-------------+----------+--------+--------+---------+---------------+
```


sample1-2.py 1
transactions.csv
question8.py 2
employee data.py 1
question8.py > ...
4 # Create Spark Session
Problems 4
Output
Debug Console
Terminal
Ports
PS D:\HCLTECH\Pyspark> python -u "d:\HCLTECH\Pyspark\tempCodeRunnerFile.py"
After Removing Null Values
+-------------+----------+--------+--------+---------+---------------+
|TransactionID|CustomerID| Product|Quantity|UnitPrice|TransactionDate|
+-------------+----------+--------+--------+---------+---------------+
4	104	Laptop	1	50000	2025-01-13
5	105	Monitor	2	10000	2025-01-14
9	109	Monitor	1	10000	2025-01-18
3	103	Keyboard	1	1500	2025-01-12
6	106	Mouse	3	500	2025-01-15
10	110	Mouse	4	500	2025-01-19
1	101	Laptop	1	50000	2025-01-10
2	102	Mouse	2	500	2025-01-11
7	107	Keyboard	2	1500	2025-01-16
8	108	Laptop	1	50000	2025-01-17
+-------------+----------+--------+--------+---------+---------------+
After Date Conversion
+-------------+----------+--------+--------+---------+---------------+
|TransactionID|CustomerID| Product|Quantity|UnitPrice|TransactionDate|
+-------------+----------+--------+--------+---------+---------------+
4	104	Laptop	1	50000	2025-01-13
5	105	Monitor	2	10000	2025-01-14
9	109	Monitor	1	10000	2025-01-18
3	103	Keyboard	1	1500	2025-01-12
6	106	Mouse	3	500	2025-01-15
10	110	Mouse	4	500	2025-01-19
1	101	Laptop	1	50000	2025-01-10
2	102	Mouse	2	500	2025-01-11
7	107	Keyboard	2	1500	2025-01-16
8	108	Laptop	1	50000	2025-01-17
+-------------+----------+--------+--------+---------+---------------+



Question 9: PySpark Joins and Performance (7 Marks)

A company maintains two datasets.

Customers

- **CustomerID**
- **CustomerName**
- **City**

Orders

- **OrderID**
- **CustomerID**
- **OrderAmount**
- **OrderDate**

Write a PySpark program to:

a. Load both datasets into DataFrames. (1 mark)

b. Join the datasets to display orders with customer details. (2 marks)

c. Identify customers who have never placed an order. (1 mark)

d. Calculate the total order value by city. (1 mark)

e. Explain when a broadcast join is useful and what conditions make it appropriate. (1 mark)



question9.py > ...

```python
order_df = spark.createDataFrame(
    orders,
    order_columns
)

print("ORDERS DATA")
order_df.show()

# b. Join Customers and Orders
print("JOIN CUSTOMERS AND ORDERS")

joined_df = customer_df.join(
    order_df,
    "CustomerID"
)

joined_df.show()

# c. Customers without Orders
print("CUSTOMERS WITHOUT ORDERS")

customer_df.join(
    order_df,
    "CustomerID",
    "left_anti"
).show()

# d. Total Order Amount by City
print("TOTAL ORDER AMOUNT BY CITY")

customer_df.join(
    order_df,
    "CustomerID"
).groupBy("City") \
 .agg(
     sum("OrderAmount").alias("TotalAmount")
 ).show()

spark.stop()
```

```
PS D:\HCLTECH\Pyspark> python -u "d:\HCLTECH\Pyspark\question9.py"
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/10/28 12:55:17 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
CUSTOMERS DATA
+----------+------------+---------+
|CustomerID|CustomerName|     City|
+----------+------------+---------+
|       101|         Ram|Bangalore|
|       102|       Ragul|  Chennai|
|       103|        Amit|   Mumbai|
|       104|        Neha|    Delhi|
|       105|        Ravi|Hyderabad|
+----------+------------+---------+

ORDERS DATA
+-------+----------+-----------+
|OrderID|CustomerID|OrderAmount|
+-------+----------+-----------+
|      1|       101|       5000|
|      2|       102|       3000|
|      3|       101|       7000|
|      4|       103|       2500|
|      5|       104|       4500|
+-------+----------+-----------+

JOIN CUSTOMERS AND ORDERS
+----------+------------+---------+-------+-----------+
|CustomerID|CustomerName|     City|OrderID|OrderAmount|
+----------+------------+---------+-------+-----------+
|       101|         Ram|Bangalore|      1|       5000|
|       101|         Ram|Bangalore|      3|       7000|
|       102|       Ragul|  Chennai|      2|       3000|
|       103|        Amit|   Mumbai|      4|       2500|
|       104|        Neha|    Delhi|      5|       4500|
+----------+------------+---------+-------+-----------+

CUSTOMERS WITHOUT ORDERS
+----------+------------+---------+
|CustomerID|CustomerName|     City|
+----------+------------+---------+
|       105|        Ravi|Hyderabad|
+----------+------------+---------+
```

```
TOTAL ORDER AMOUNT BY CITY
+---------+-----------+
|     City|TotalAmount|
+---------+-----------+
|Bangalore|      12000|
|  Chennai|       3000|
|   Mumbai|       2500|
|    Delhi|       4500|
+---------+-----------+

PS D:\HCLTECH\Pyspark> python -u "d:\HCLTECH\Pyspark\question9.py"
SUCCESS: The process with PID 26792 (child process of PID 21156) has been terminated.
SUCCESS: The process with PID 21156 (child process of PID 23856) has been terminated.
SUCCESS: The process with PID 23856 (child process of PID 17896) has been terminated.
WARNING: Using incubator modules: jdk.incubator.vector
```

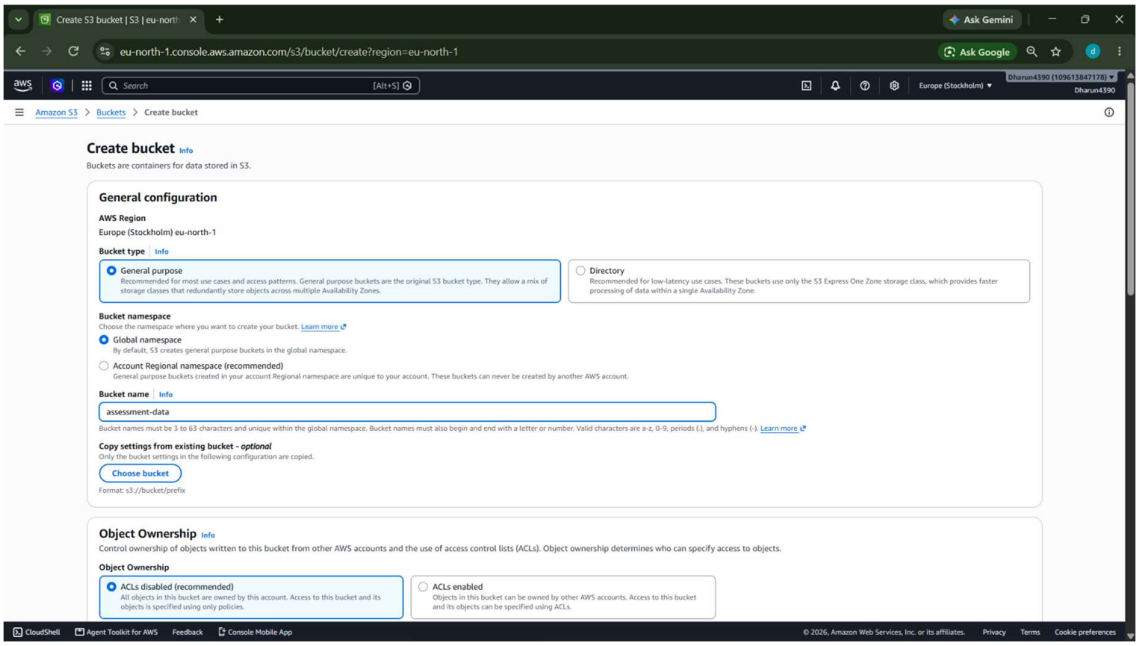
SECTION D: AWS CLOUD SERVICES || Maximum Marks: 15

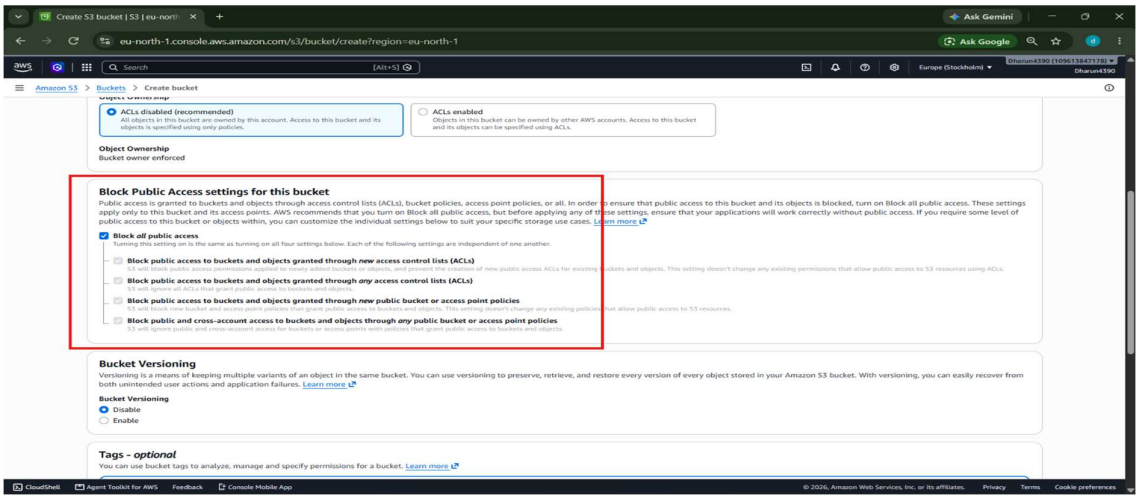
Question 10: S3 and IAM Security (5 Marks)

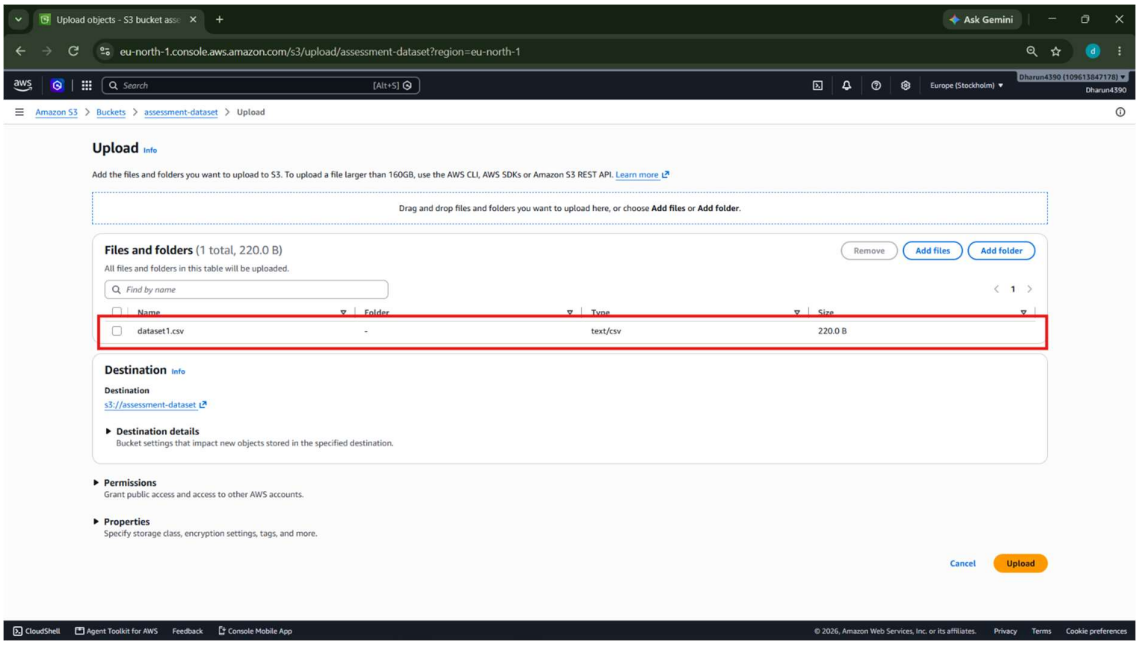
A company stores confidential employee reports in Amazon S3. Only authorized HR employees should be able to read these reports.

Perform or document the following:

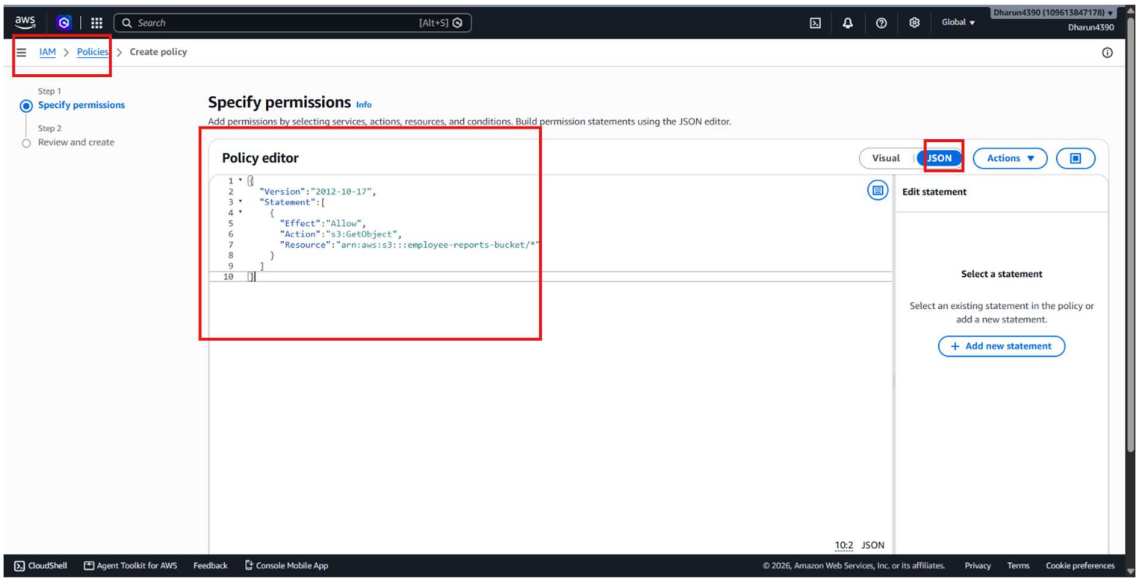
a. Create an S3 bucket with Block Public Access enabled and upload a sample file. (1 mark)

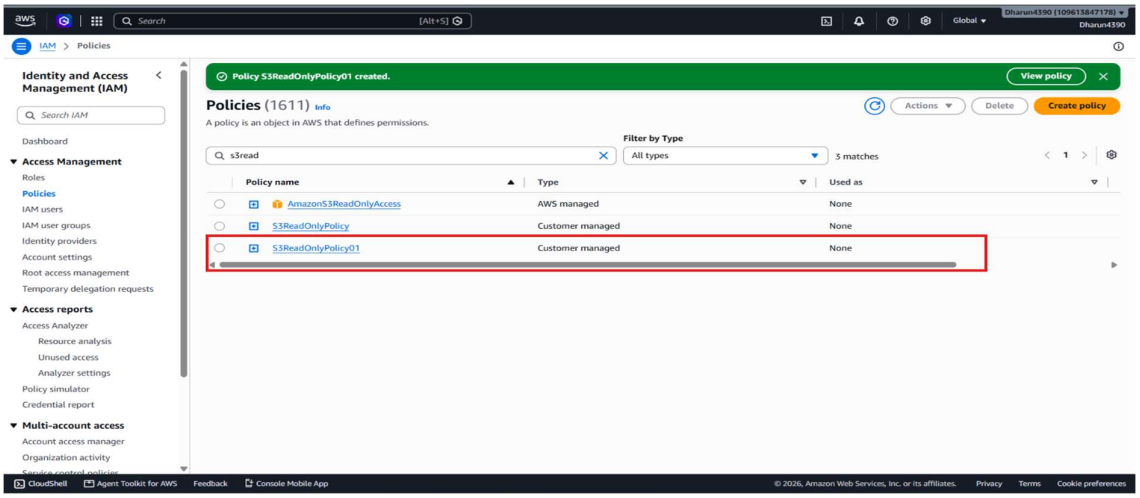




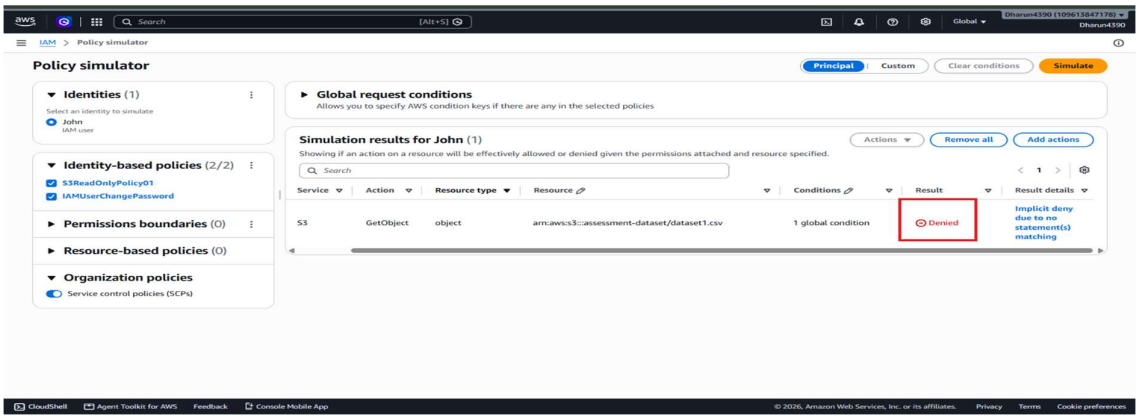


b. Design an IAM policy that grants read access to objects in the designated bucket without granting access to unrelated buckets. (2 marks)

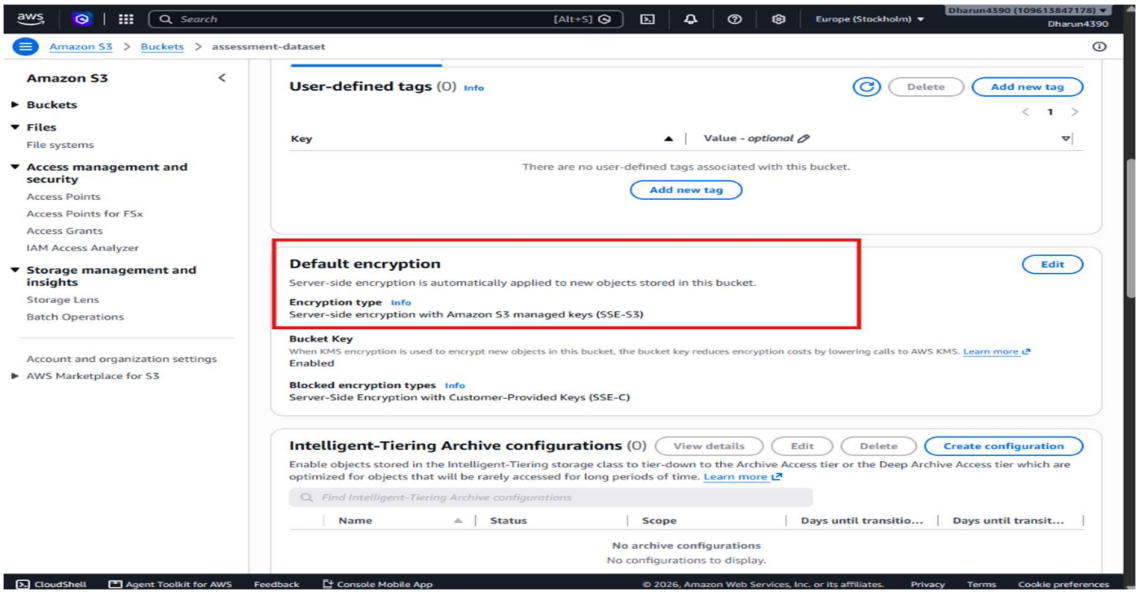




c. Explain how you would verify that an unauthorized user cannot read the file. (1 mark)



d. Explain how encryption at rest protects the stored data. (1 mark)

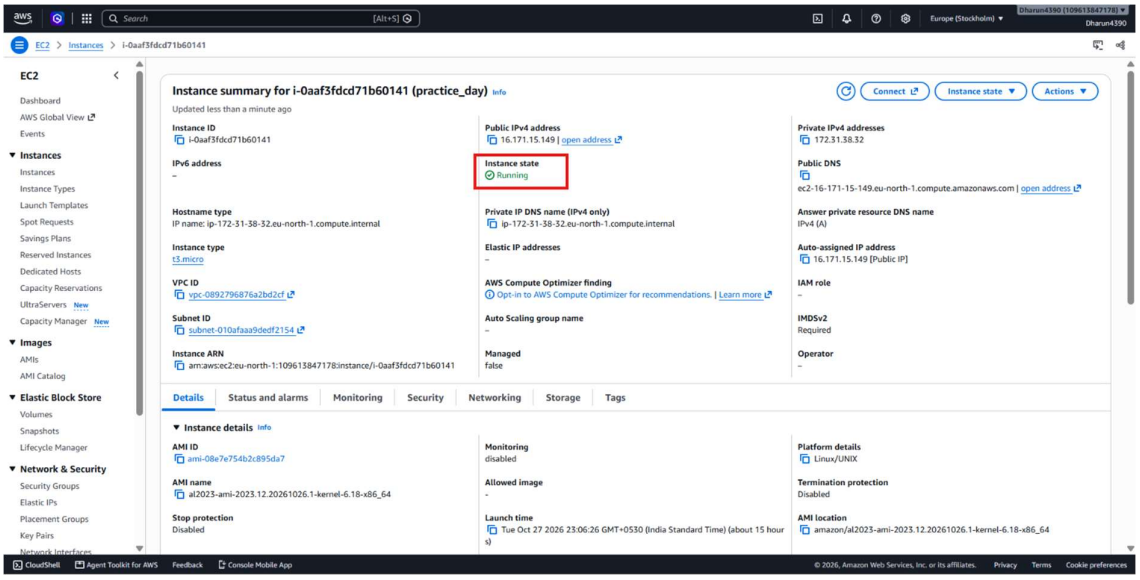


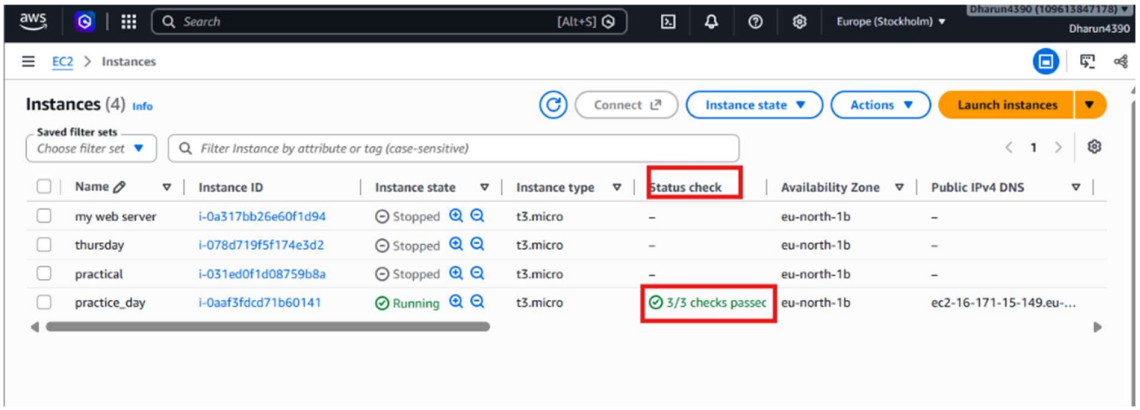
Question 11: EC2 Troubleshooting and CloudWatch (5 Marks)

A web application hosted on EC2 becomes slow, and users report connectivity problems.

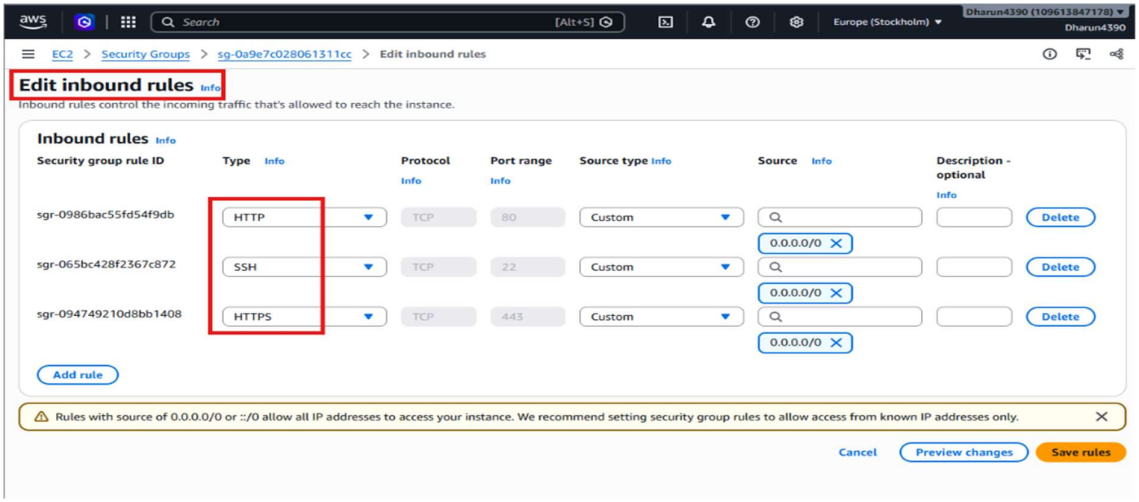
Perform or document the following:

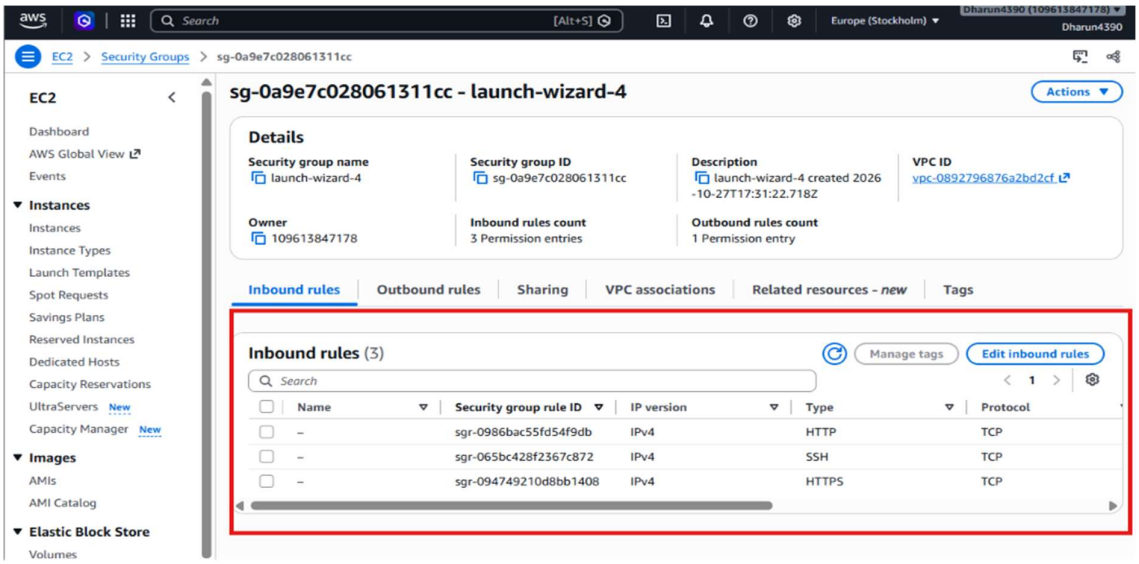
a. Check the EC2 instance state and status checks. (1 mark)

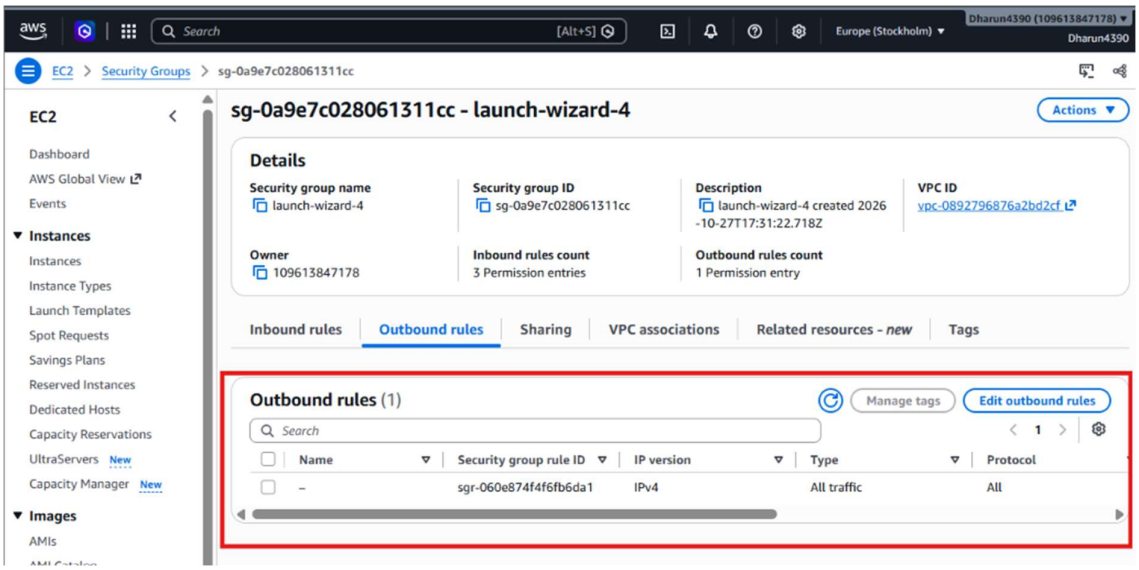


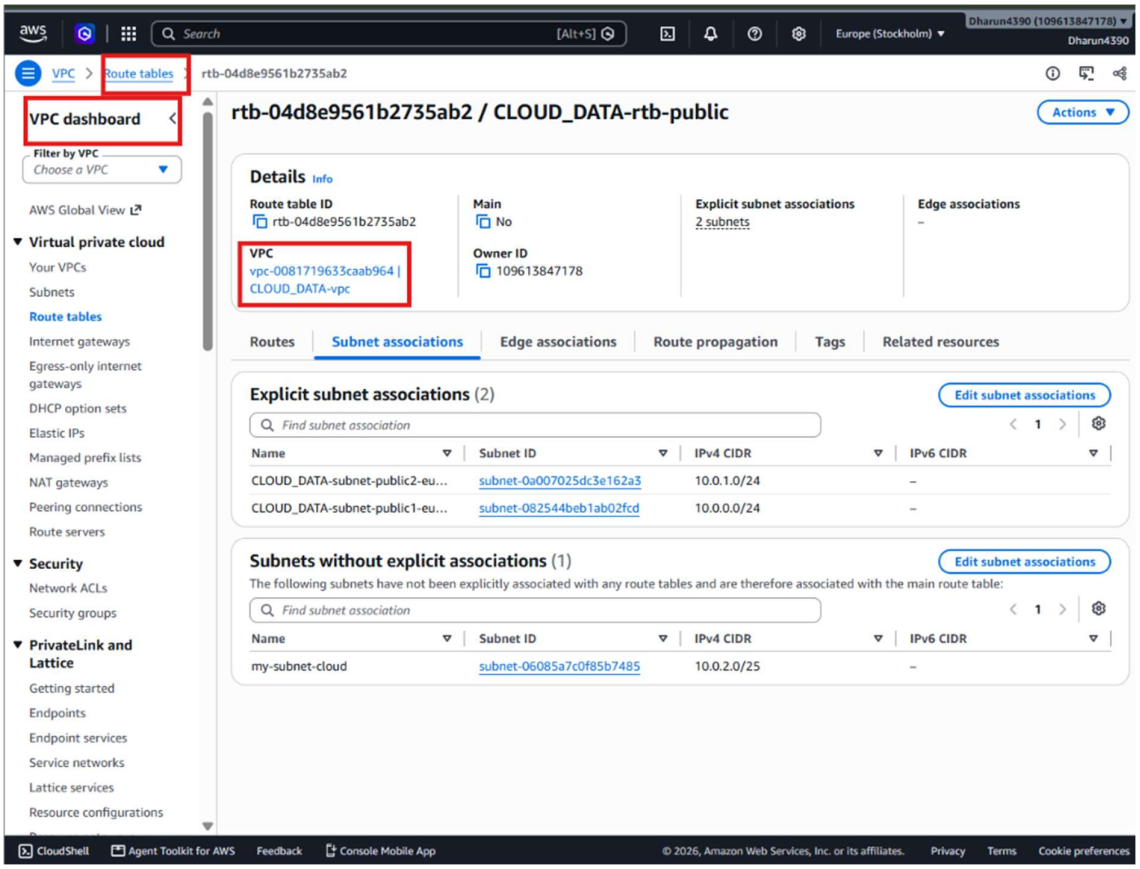


b. Explain how security groups and route tables can affect application connectivity. (1 mark)

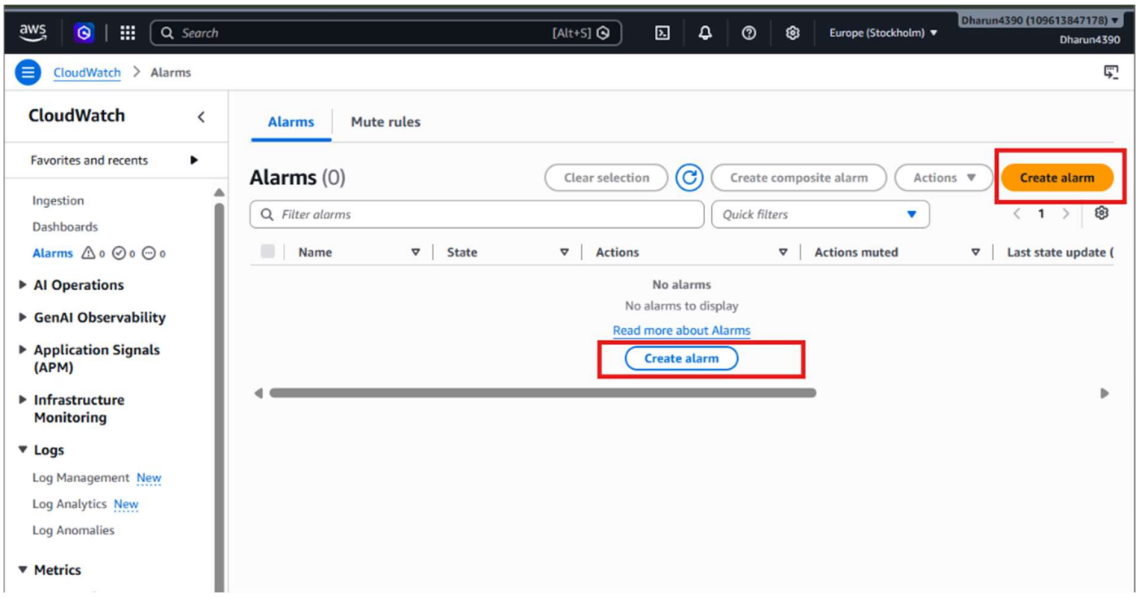

Edit inbound rules
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules
Security group rule ID
Type
Protocol
Port range
Source type
Source
Description - optional
sgr-0986bac55fd54f9db
HTTP
TCP
80
Custom
0.0.0.0/0
Delete
sgr-065bc428f2367c872
SSH
TCP
22
Custom
0.0.0.0/0
Delete
sgr-094749210d8bb1408
HTTPS
TCP
443
Custom
0.0.0.0/0
Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Cancel
Preview changes
Save rules

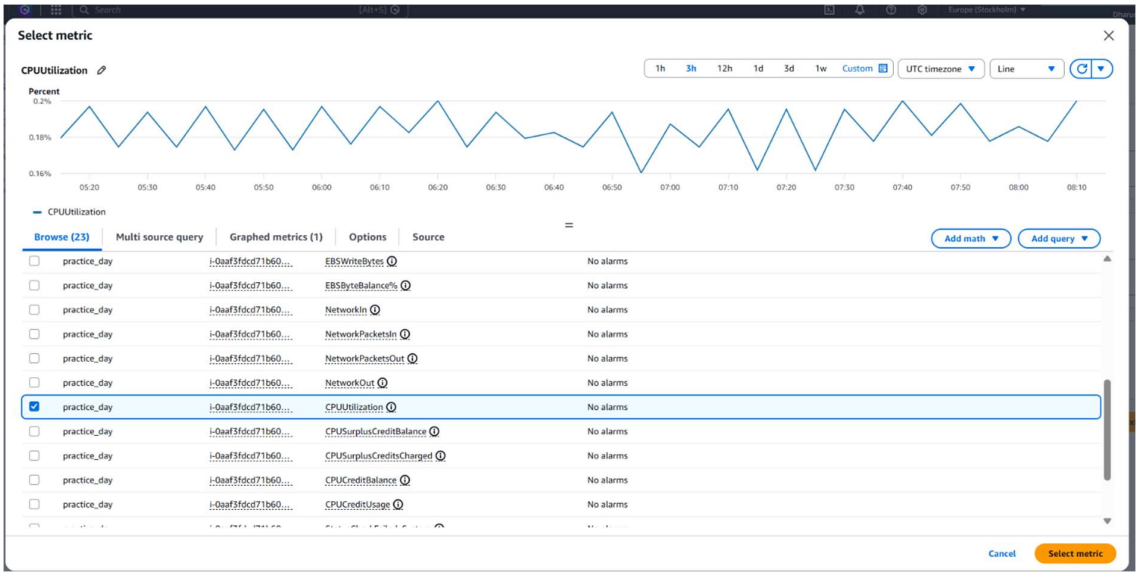

sg-0a9e7c028061311cc - launch-wizard-4
Actions
Details
Security group name
launch-wizard-4
Security group ID
sg-0a9e7c028061311cc
Description
launch-wizard-4 created 2026-10-27T17:31:22.718Z
VPC ID
vpc-0892796876a2bd2cf
Owner
109613847178
Inbound rules count
3 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (3)
Manage tags
Edit inbound rules
Name
Security group rule ID
IP version
Type
Protocol
sgr-0986bac55fd54f9db
IPv4
HTTP
TCP
sgr-065bc428f2367c872
IPv4
SSH
TCP
sgr-094749210d8bb1408
IPv4
HTTPS
TCP

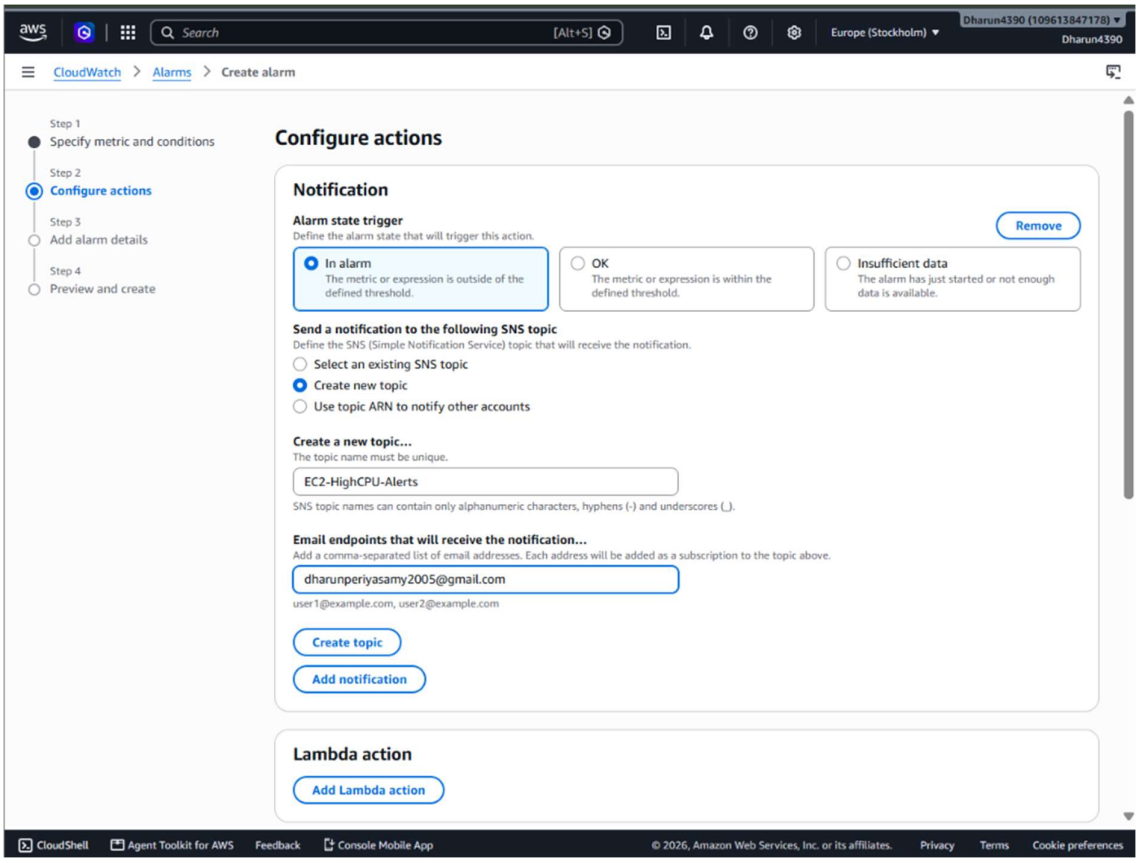

sg-0a9e7c028061311cc - launch-wizard-4
Actions
Details
Security group name
launch-wizard-4
Security group ID
sg-0a9e7c028061311cc
Description
launch-wizard-4 created 2026-10-27T17:31:22.718Z
VPC ID
vpc-0892796876a2bd2cf
Owner
109613847178
Inbound rules count
3 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Outbound rules (1)
Manage tags
Edit outbound rules
Name
Security group rule ID
IP version
Type
Protocol
sgr-060e874f4f6fb6da1
IPv4
All traffic
All

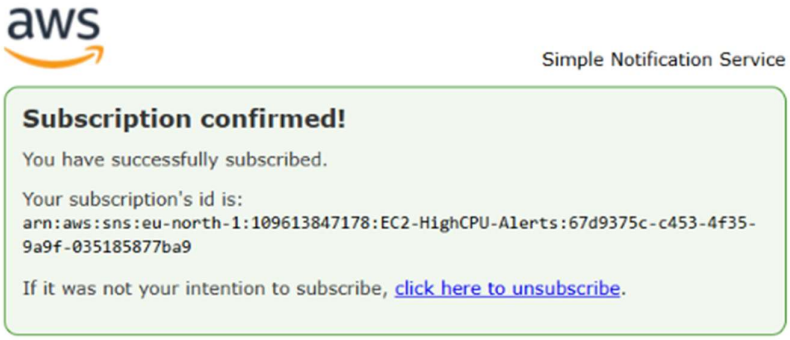


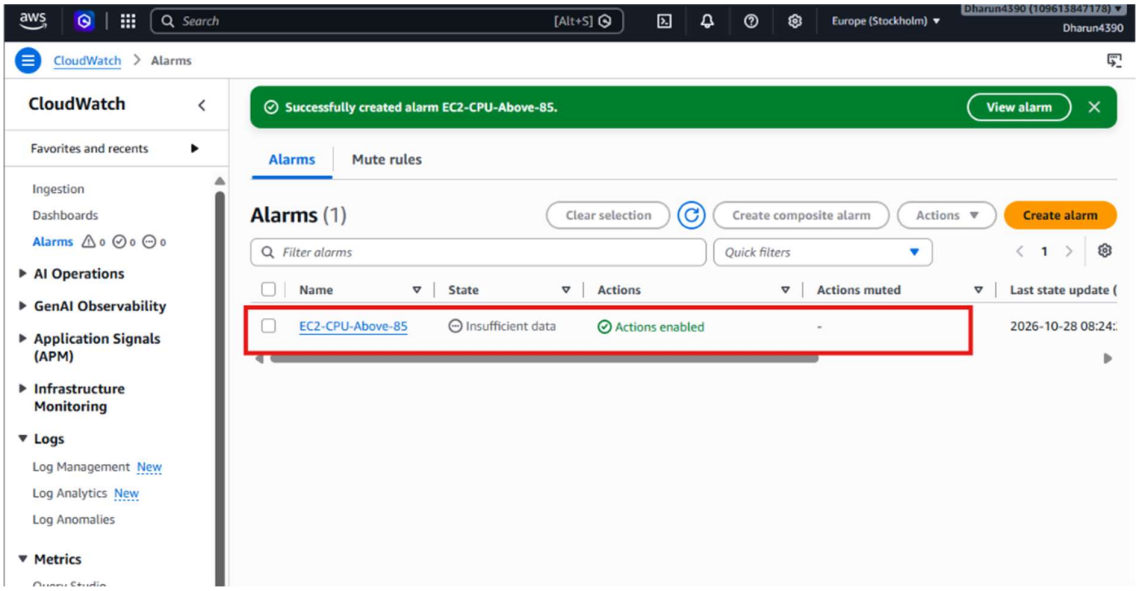
c. Explain how to configure a CloudWatch alarm when CPU utilization exceeds 85% for a defined evaluation period. (2 marks)



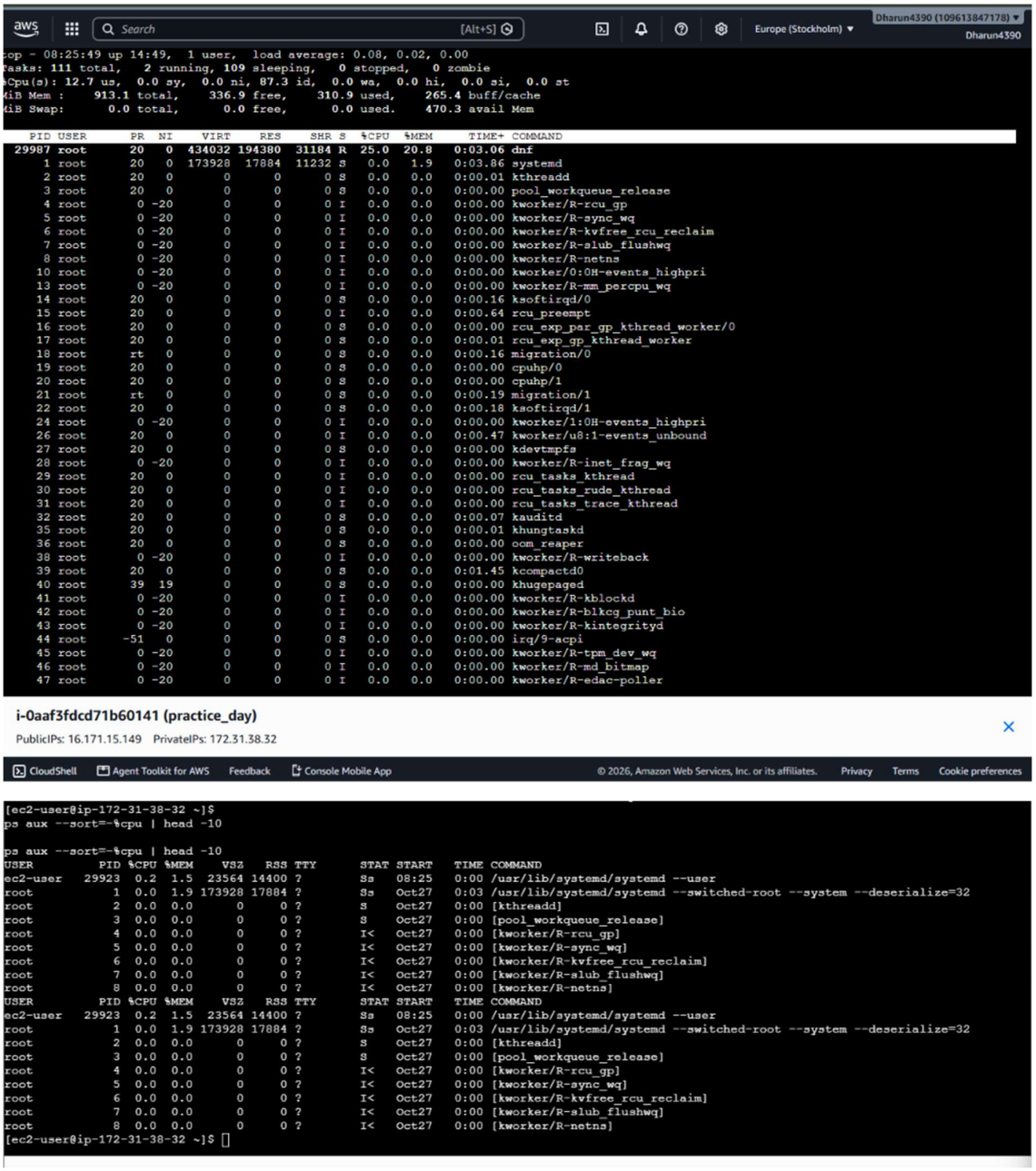

Select metric
CPUUtilization
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Line
Percent
0.2%
0.18%
0.16%
05:20 05:30 05:40 05:50 06:00 06:10 06:20 06:30 06:40 06:50 07:00 07:10 07:20 07:30 07:40 07:50 08:00 08:10
CPUUtilization
Browse (23)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
practice_day i-0aaf3fdcd71b60... EBSWriteBytes No alarms
practice_day i-0aaf3fdcd71b60... EBSByteBalance% No alarms
practice_day i-0aaf3fdcd71b60... NetworkIn No alarms
practice_day i-0aaf3fdcd71b60... NetworkPacketsIn No alarms
practice_day i-0aaf3fdcd71b60... NetworkPacketsOut No alarms
practice_day i-0aaf3fdcd71b60... NetworkOut No alarms
practice_day i-0aaf3fdcd71b60... CPUUtilization No alarms
practice_day i-0aaf3fdcd71b60... CPUSurplusCreditBalance No alarms
practice_day i-0aaf3fdcd71b60... CPUSurplusCreditsCharged No alarms
practice_day i-0aaf3fdcd71b60... CPUCreditBalance No alarms
practice_day i-0aaf3fdcd71b60... CPUCreditUsage No alarms
Cancel
Select metric


Search [Alt+S]
Europe (Stockholm)
Dharun4390 (109613847178)
Dharun4390
CloudWatch > Alarms > Create alarm
Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Configure actions
Notification
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.
Select an existing SNS topic
Create new topic
Use topic ARN to notify other accounts
Create a new topic...
The topic name must be unique.
EC2-HighCPU-Alerts
SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).
Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.
dharunperiyasamy2005@gmail.com
user1@example.com, user2@example.com
Create topic
Add notification
Lambda action
Add Lambda action
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences





d. Identify one method for investigating the root cause of high CPU utilization. (1 mark)



Question 12: Event-Driven Processing (5 Marks)

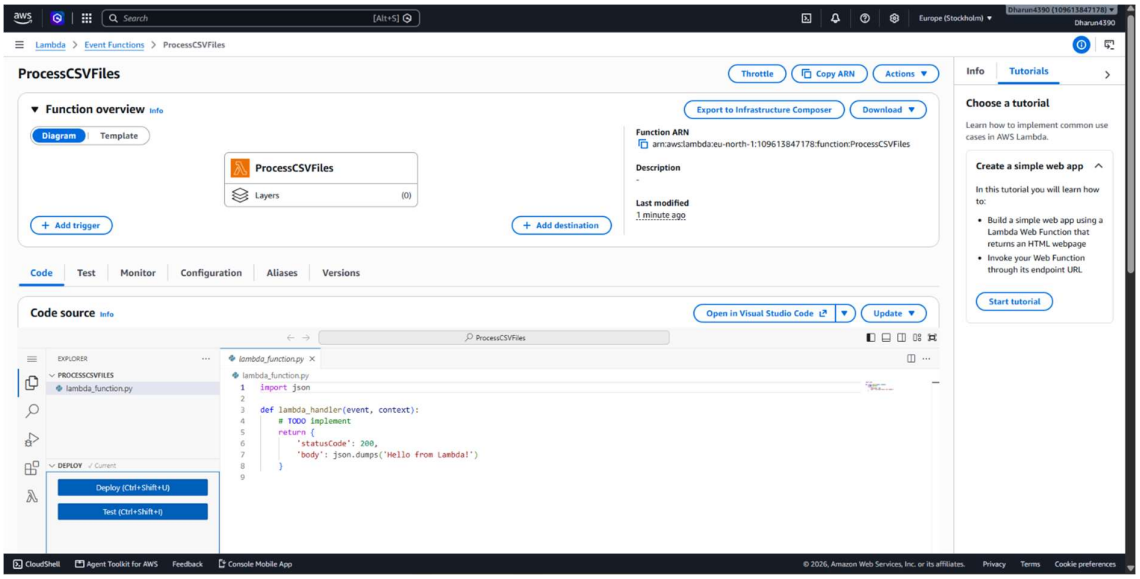
A company uploads CSV files to an S3 bucket.

Each uploaded file must be validated and processed automatically, with the results stored in another location.

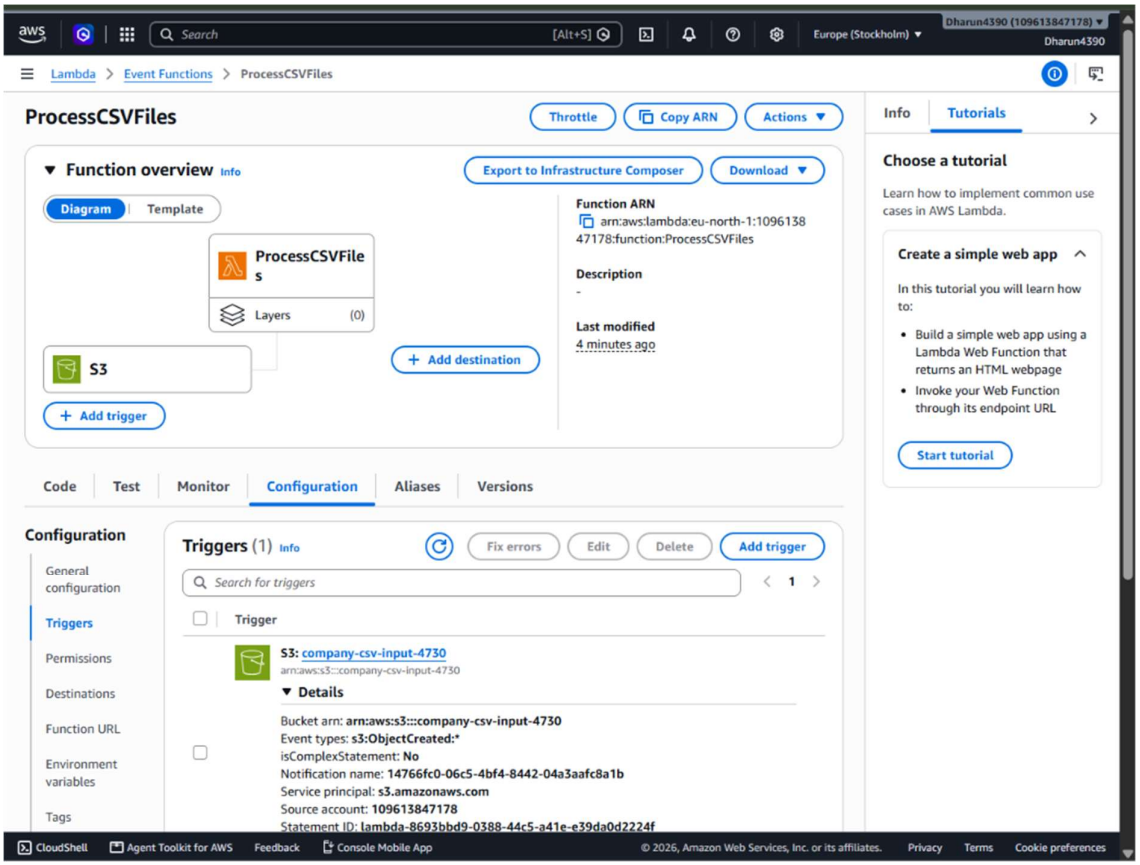
Design a solution using AWS services.

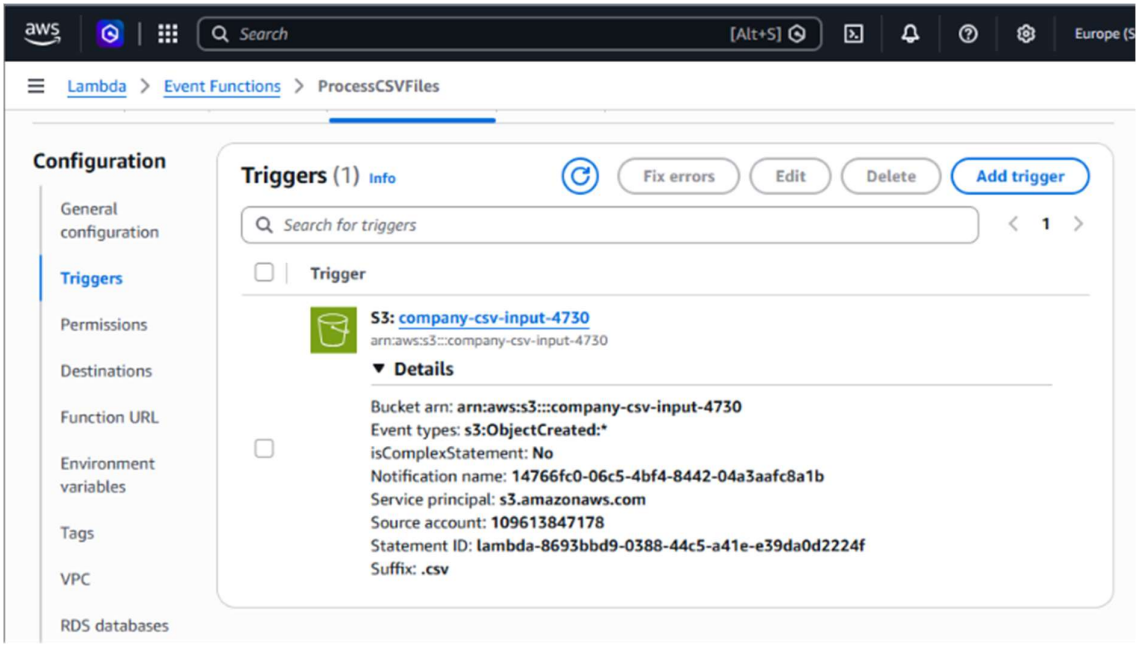
a. Identify a suitable service for executing the processing code. (1 mark)

Create an AWS Lambda function that will execute Python code whenever a CSV file is uploaded to Amazon S3.

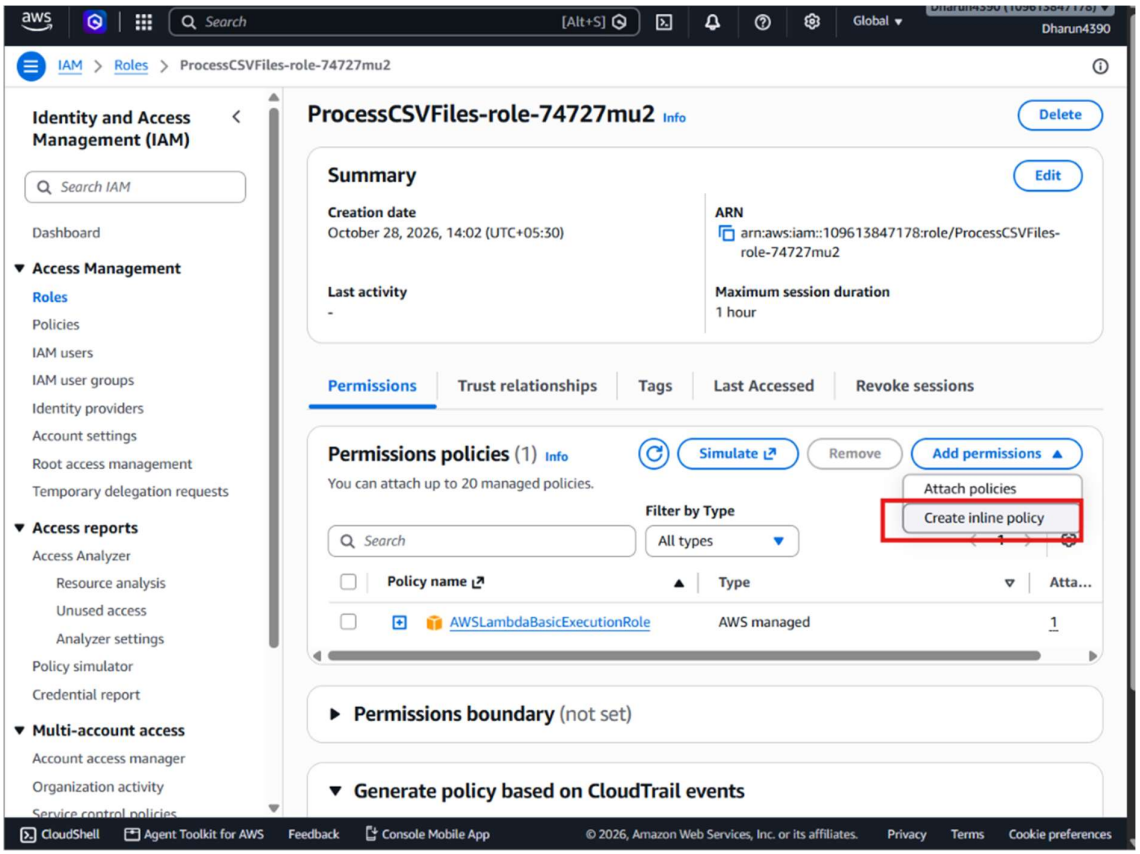


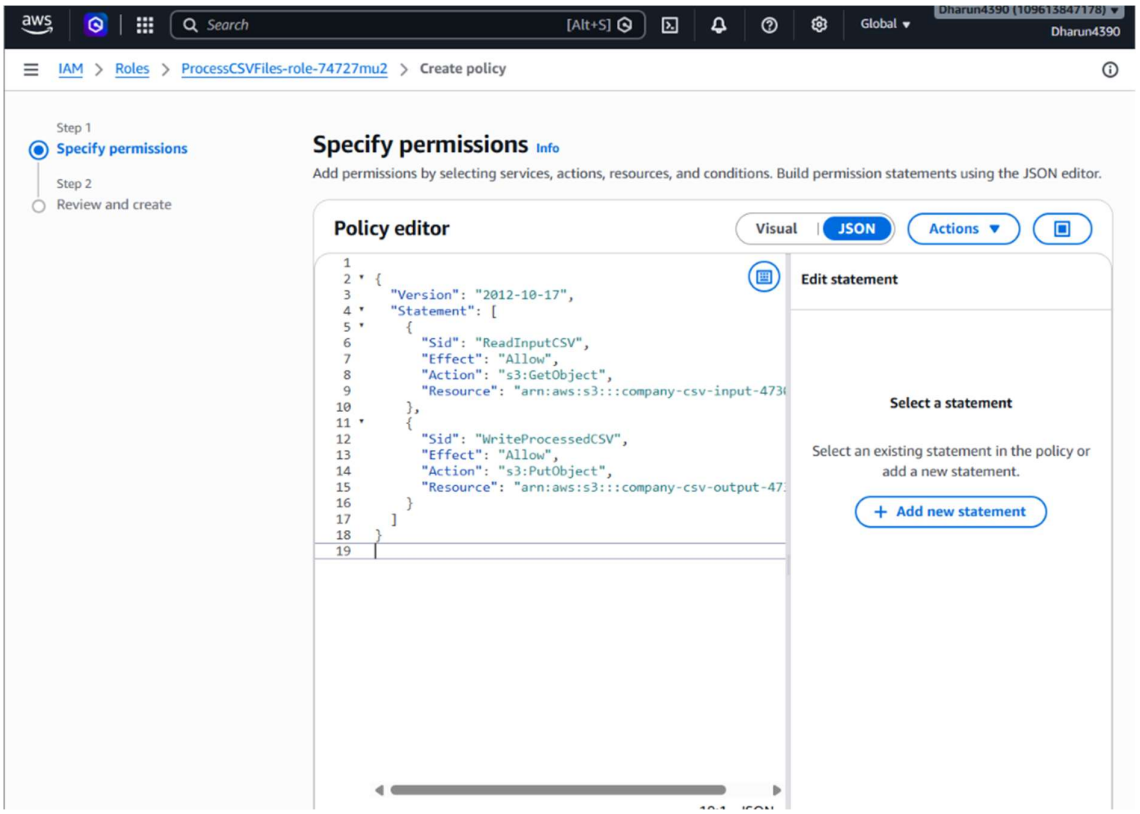
b. Explain how an S3 upload can trigger processing. (1 mark)

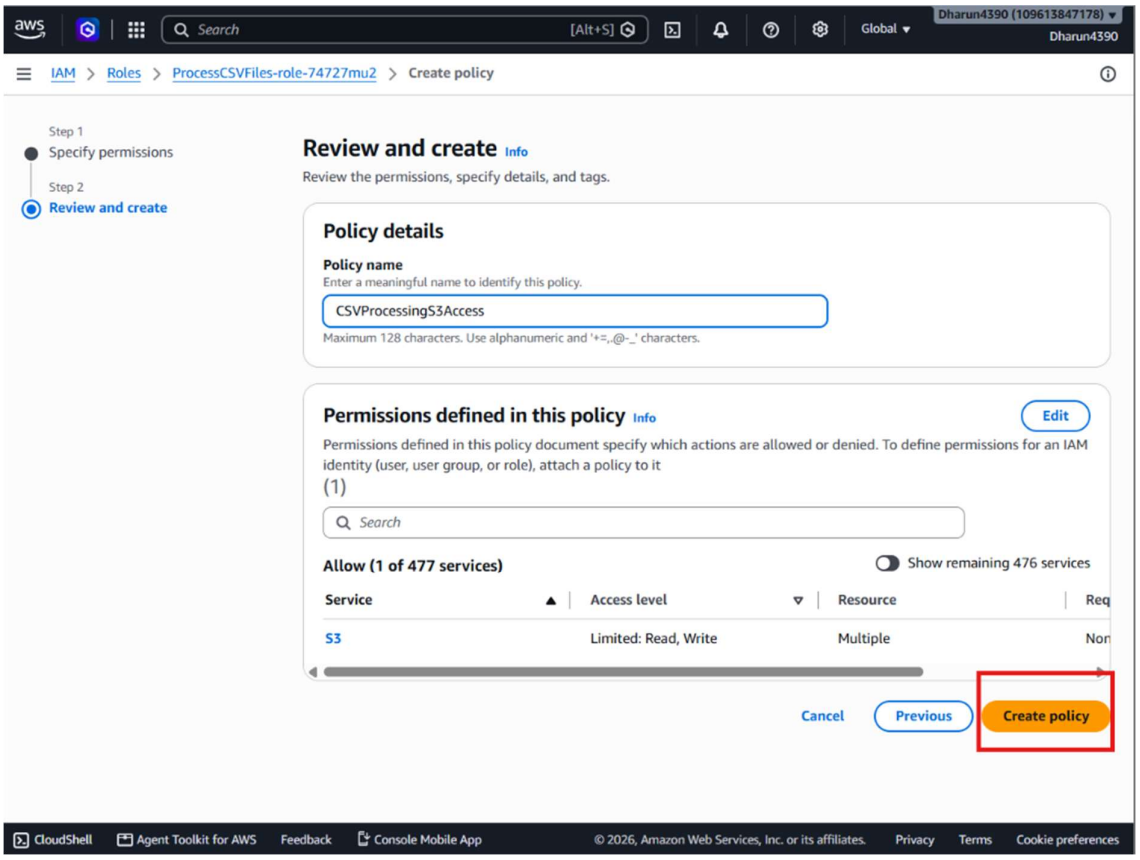


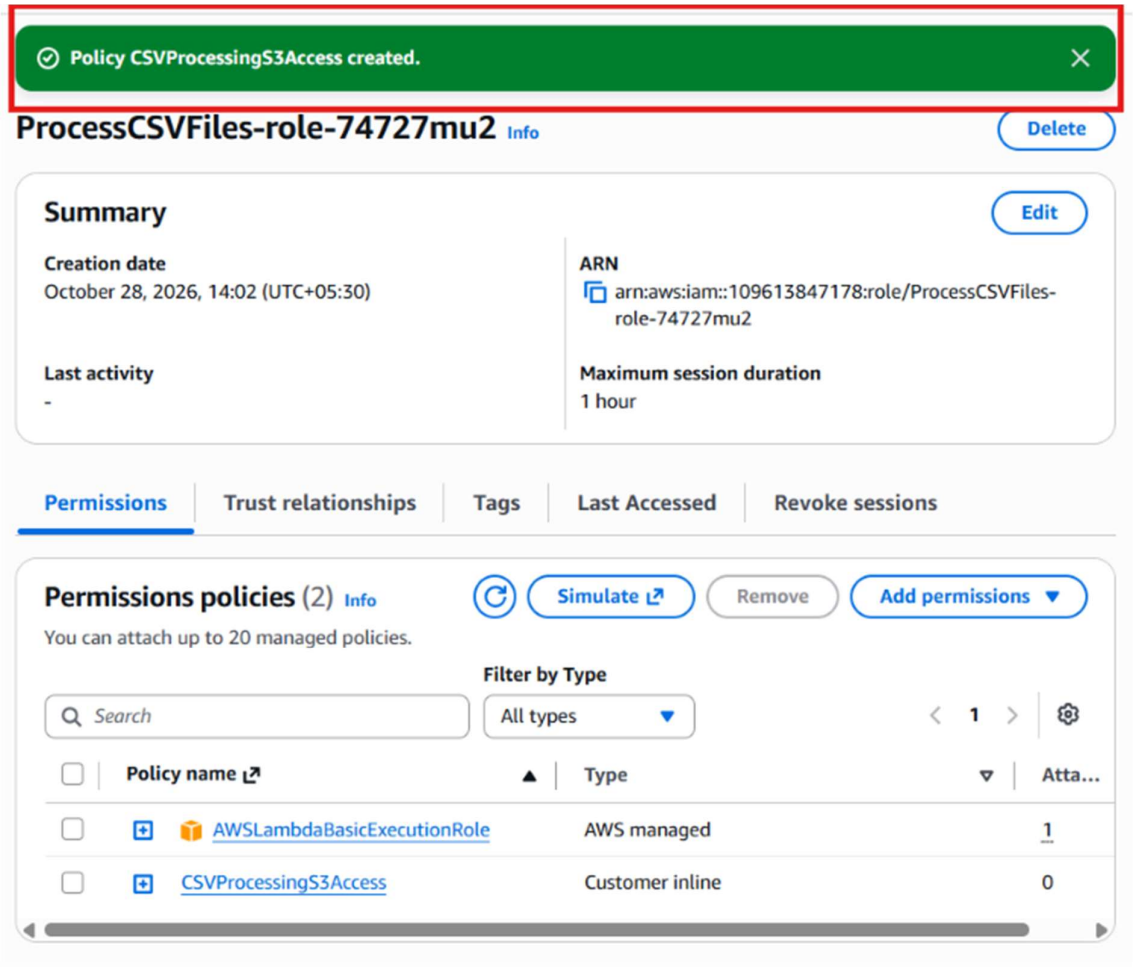


c. Explain the IAM permissions needed to read the input file and write the results. (1 mark)

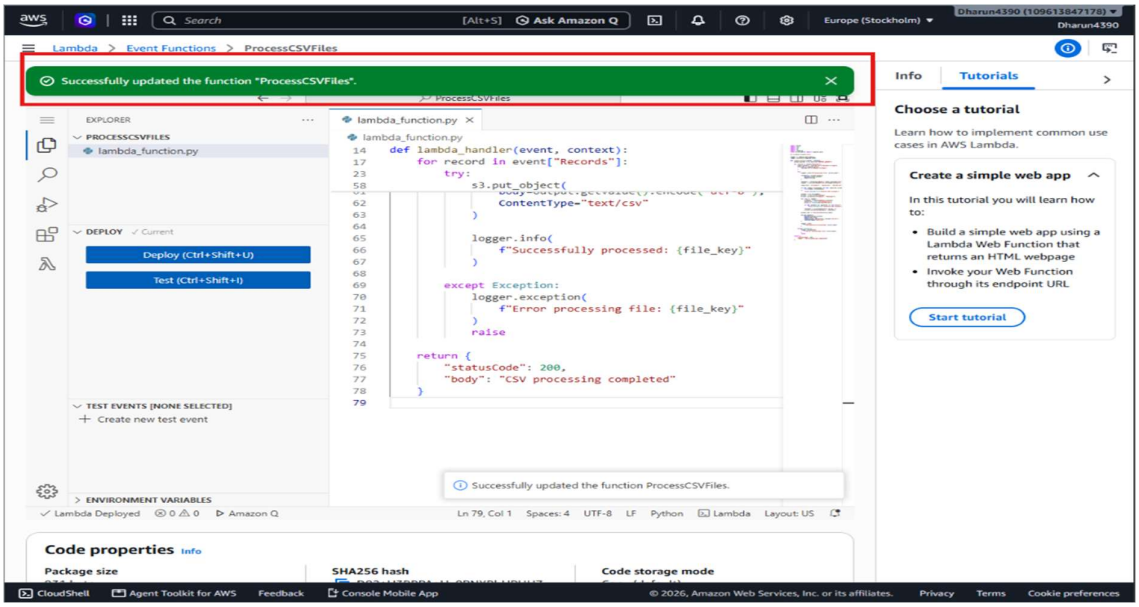


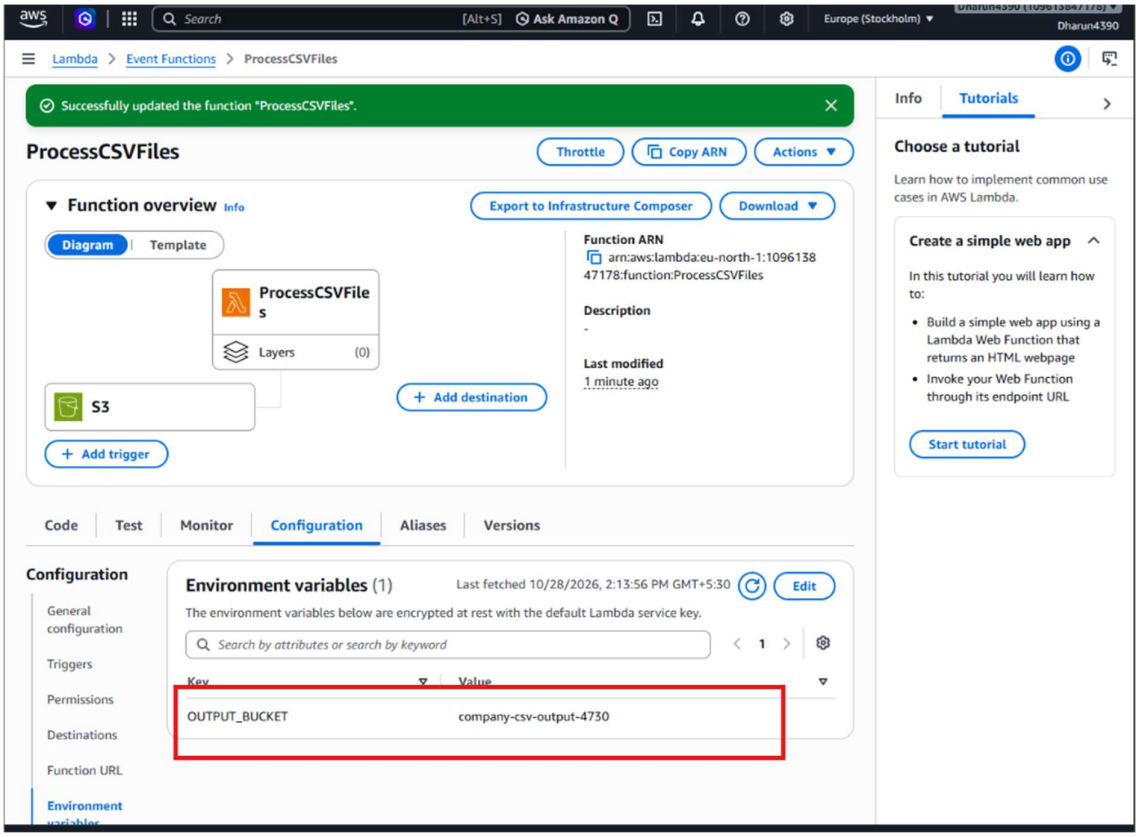

Search
[Alt+S]
Global
Dharun4390 (109613847178)
Dharun4390
IAM > Roles > ProcessCSVFiles-role-74727mu2 > Create policy
Step 1
Specify permissions
Step 2
Review and create
Specify permissions Info
Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.
Policy editor
Visual
JSON
Actions
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "ReadInputCSV",
"Effect": "Allow",
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::company-csv-input-473
},
{
"Sid": "WriteProcessedCSV",
"Effect": "Allow",
"Action": "s3:PutObject",
"Resource": "arn:aws:s3:::company-csv-output-47
}
]
}
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
+ Add new statement

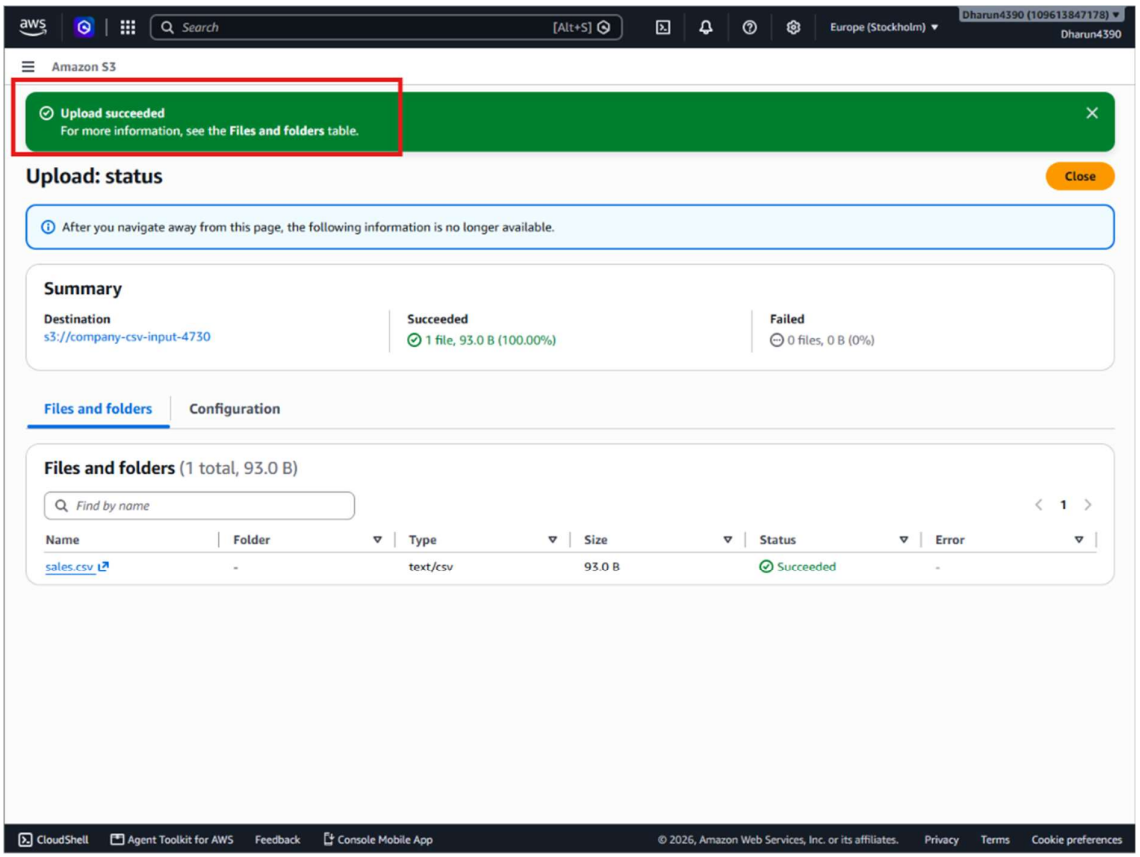

Search
[Alt+S]
Global
Dharun4390 (109613847178)
Dharun4390
IAM > Roles > ProcessCSVFiles-role-74727mu2 > Create policy
Step 1
Specify permissions
Step 2
Review and create
Review and create Info
Review the permissions, specify details, and tags.
Policy details
Policy name
Enter a meaningful name to identify this policy.
CSVProcessingS3Access
Maximum 128 characters. Use alphanumeric and '+=,.@-_' characters.
Permissions defined in this policy Info
Edit
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it
(1)
Search
Allow (1 of 477 services)
Show remaining 476 services
Service
Access level
Resource
Req
S3
Limited: Read, Write
Multiple
Non
Cancel
Previous
Create policy
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

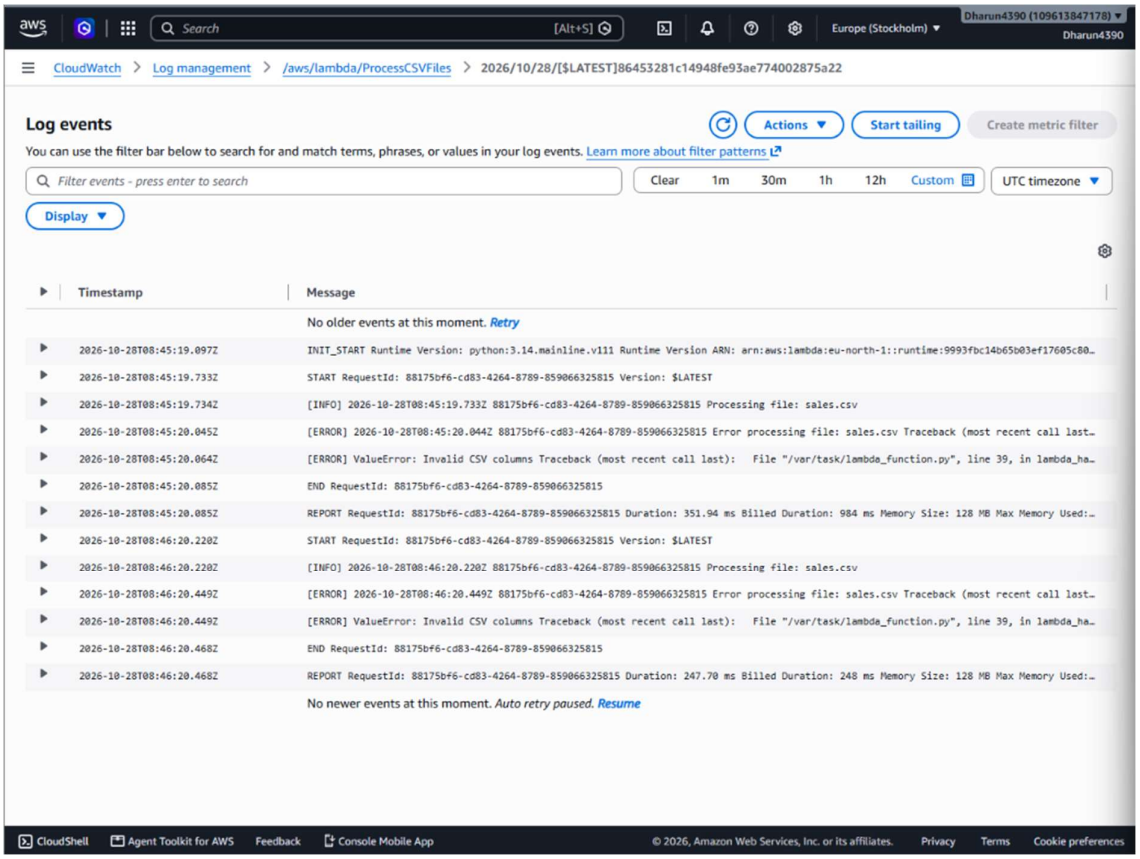


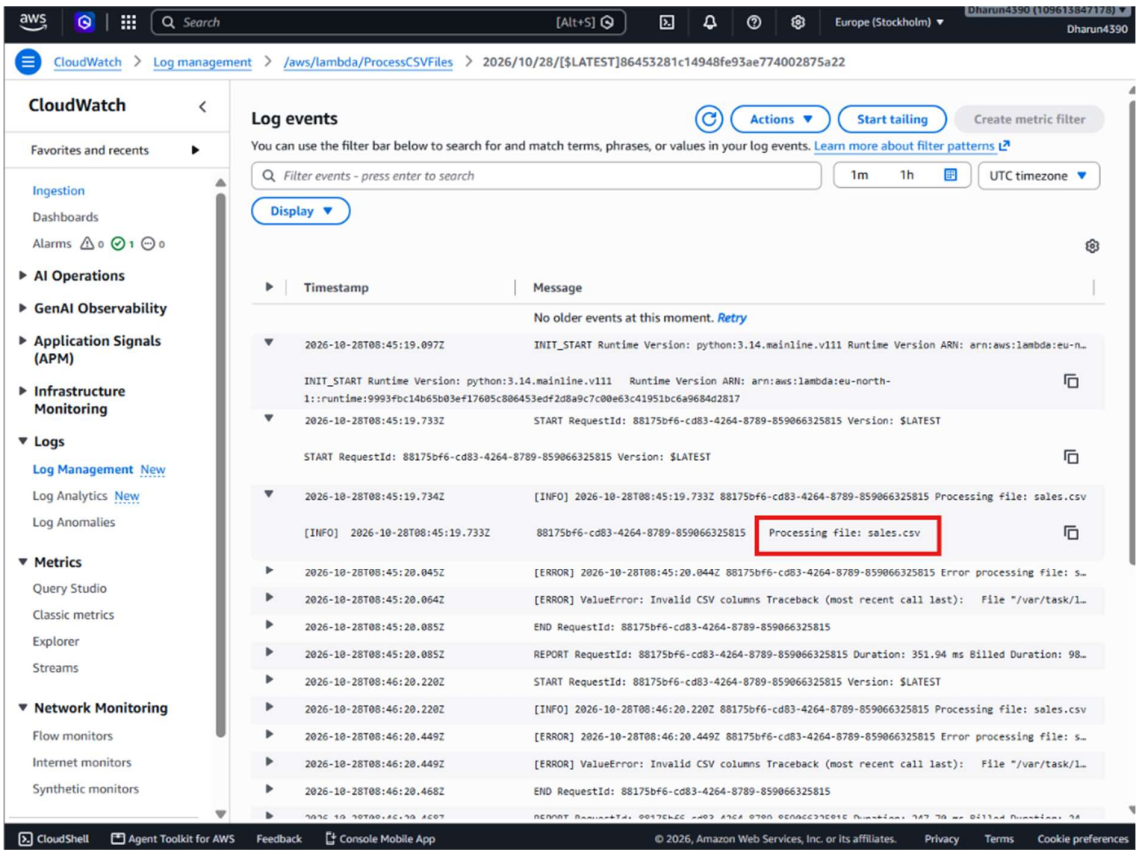
d. Explain how errors and processing logs can be captured. (1 mark)



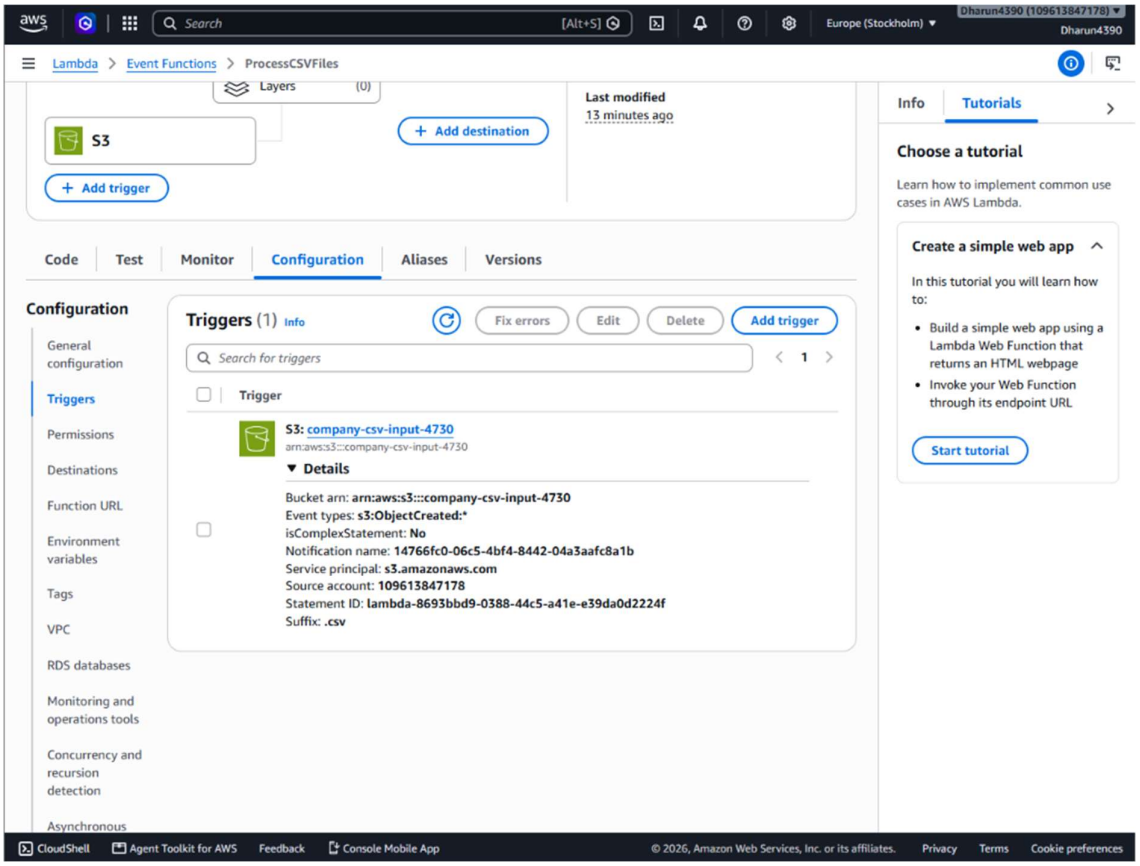

Lambda > Event Functions > ProcessCSVFiles
Successfully updated the function "ProcessCSVFiles".
ProcessCSVFiles
Throttle
Copy ARN
Actions
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
ProcessCSVFiles
Layers (0)
S3
Add destination
Add trigger
Function ARN
arn:aws:lambda:eu-north-1:109613847178:function:ProcessCSVFiles
Description
-
Last modified
1 minute ago
Info
Tutorials
Choose a tutorial
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app using a Lambda Web Function that returns an HTML webpage
Invoke your Web Function through its endpoint URL
Start tutorial
Code
Test
Monitor
Configuration
Aliases
Versions
Configuration
General configuration
Triggers
Permissions
Destinations
Function URL
Environment variables
Environment variables (1)
Last fetched 10/28/2026, 2:13:56 PM GMT+5:30
Edit
The environment variables below are encrypted at rest with the default Lambda service key.
Search by attributes or search by keyword
Key
Value
OUTPUT_BUCKET
company-csv-output-4730


Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://company-csv-input-4730
Succeeded
1 file, 93.0 B (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 93.0 B)
Find by name
Name
Folder
Type
Size
Status
Error
sales.csv
-
text/csv
93.0 B
Succeeded
-
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Search [Alt+S]
Europe (Stockholm)
Dharun4390 (109613847178)
Dharun4390
CloudWatch > Log management > /aws/lambda/ProcessCSVFiles > 2026/10/28/[$LATEST]86453281c14948fe93ae774002875a22
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear 1m 30m 1h 12h Custom
UTC timezone
Display
Timestamp
Message
No older events at this moment. Retry
2026-10-28T08:45:19.097Z INIT_START Runtime Version: python:3.14.mainline.v111 Runtime Version ARN: arn:aws:lambda:eu-north-1::runtime:9993fbc14b65b03ef17605c80…
2026-10-28T08:45:19.733Z START RequestId: 88175bf6-cd83-4264-8789-859066325815 Version: $LATEST
2026-10-28T08:45:19.734Z [INFO] 2026-10-28T08:45:19.733Z 88175bf6-cd83-4264-8789-859066325815 Processing file: sales.csv
2026-10-28T08:45:20.045Z [ERROR] 2026-10-28T08:45:20.044Z 88175bf6-cd83-4264-8789-859066325815 Error processing file: sales.csv Traceback (most recent call last…
2026-10-28T08:45:20.064Z [ERROR] ValueError: Invalid CSV columns Traceback (most recent call last): File "/var/task/lambda_function.py", line 39, in lambda_ha…
2026-10-28T08:45:20.085Z END RequestId: 88175bf6-cd83-4264-8789-859066325815
2026-10-28T08:45:20.085Z REPORT RequestId: 88175bf6-cd83-4264-8789-859066325815 Duration: 351.94 ms Billed Duration: 984 ms Memory Size: 128 MB Max Memory Used:…
2026-10-28T08:46:20.220Z START RequestId: 88175bf6-cd83-4264-8789-859066325815 Version: $LATEST
2026-10-28T08:46:20.220Z [INFO] 2026-10-28T08:46:20.220Z 88175bf6-cd83-4264-8789-859066325815 Processing file: sales.csv
2026-10-28T08:46:20.449Z [ERROR] 2026-10-28T08:46:20.449Z 88175bf6-cd83-4264-8789-859066325815 Error processing file: sales.csv Traceback (most recent call last…
2026-10-28T08:46:20.449Z [ERROR] ValueError: Invalid CSV columns Traceback (most recent call last): File "/var/task/lambda_function.py", line 39, in lambda_ha…
2026-10-28T08:46:20.468Z END RequestId: 88175bf6-cd83-4264-8789-859066325815
2026-10-28T08:46:20.468Z REPORT RequestId: 88175bf6-cd83-4264-8789-859066325815 Duration: 247.70 ms Billed Duration: 248 ms Memory Size: 128 MB Max Memory Used:…
No newer events at this moment. Auto retry paused. Resume
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CloudWatch > Log management > /aws/lambda/ProcessCSVFiles > 2026/10/28/[$LATEST]86453281c14948fe93ae774002875a22
CloudWatch
Favorites and recents
Ingestion
Dashboards
Alarms 0 1 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management New
Log Analytics New
Log Anomalies
Metrics
Query Studio
Classic metrics
Explorer
Streams
Network Monitoring
Flow monitors
Internet monitors
Synthetic monitors
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
1m 1h
UTC timezone
Display
Timestamp
Message
No older events at this moment. Retry
2026-10-28T08:45:19.097Z INIT_START Runtime Version: python:3.14.mainline.v111 Runtime Version ARN: arn:aws:lambda:eu-n…
INIT_START Runtime Version: python:3.14.mainline.v111 Runtime Version ARN: arn:aws:lambda:eu-north-1::runtime:9993fbc14b65b03ef17605c806453edf2d8a9c7c00e63c41951bc6a9684d2817
2026-10-28T08:45:19.733Z START RequestId: 88175bf6-cd83-4264-8789-859066325815 Version: $LATEST
START RequestId: 88175bf6-cd83-4264-8789-859066325815 Version: $LATEST
2026-10-28T08:45:19.734Z [INFO] 2026-10-28T08:45:19.733Z 88175bf6-cd83-4264-8789-859066325815 Processing file: sales.csv
[INFO] 2026-10-28T08:45:19.733Z 88175bf6-cd83-4264-8789-859066325815 Processing file: sales.csv
2026-10-28T08:45:20.045Z [ERROR] 2026-10-28T08:45:20.044Z 88175bf6-cd83-4264-8789-859066325815 Error processing file: s…
2026-10-28T08:45:20.064Z [ERROR] ValueError: Invalid CSV columns Traceback (most recent call last): File "/var/task/l…
2026-10-28T08:45:20.085Z END RequestId: 88175bf6-cd83-4264-8789-859066325815
2026-10-28T08:45:20.085Z REPORT RequestId: 88175bf6-cd83-4264-8789-859066325815 Duration: 98…
2026-10-28T08:46:20.220Z START RequestId: 88175bf6-cd83-4264-8789-859066325815 Version: $LATEST
2026-10-28T08:46:20.220Z [INFO] 2026-10-28T08:46:20.220Z 88175bf6-cd83-4264-8789-859066325815 Processing file: sales.csv
2026-10-28T08:46:20.449Z [ERROR] 2026-10-28T08:46:20.449Z 88175bf6-cd83-4264-8789-859066325815 Error processing file: s…
2026-10-28T08:46:20.449Z [ERROR] ValueError: Invalid CSV columns Traceback (most recent call last): File "/var/task/l…
2026-10-28T08:46:20.468Z END RequestId: 88175bf6-cd83-4264-8789-859066325815
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e. Explain how to prevent duplicate processing or recursive triggering. (1 mark)



SECTION E: AWS STEP FUNCTIONS || Maximum Marks: 15

Question 13: State Machine Design (7 Marks)

A company needs to process sales files through the following stages:

1. Validate the input file.

2. Transform the data.

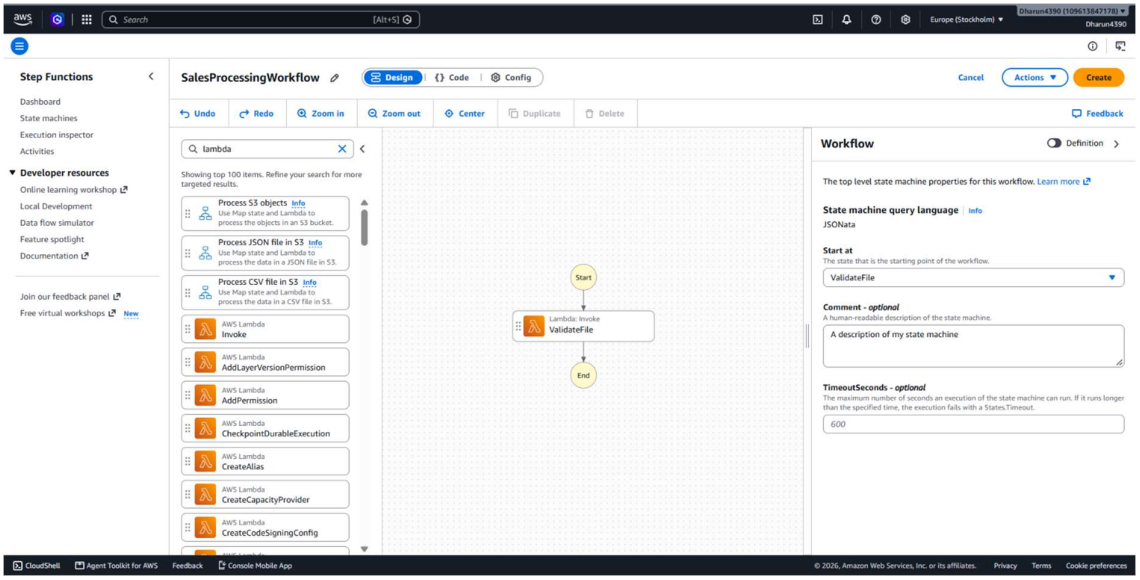
3. Calculate total sales by product.

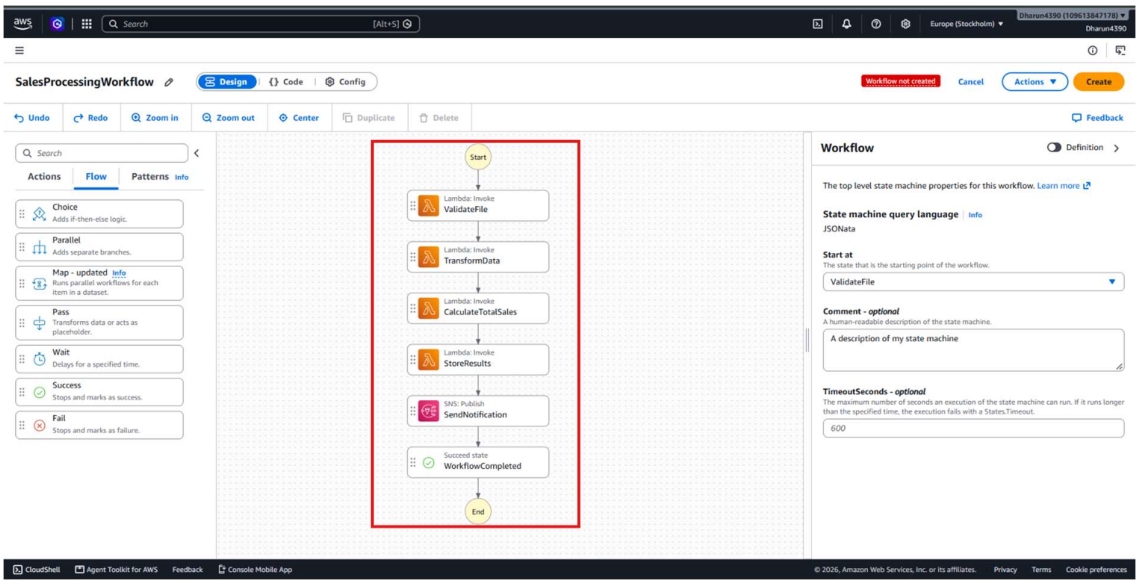
4. Store the results in Amazon S3.

5. Send a completion notification.

Design an AWS Step Functions state machine to coordinate the workflow.

a. Identify suitable states for validation, processing, storage, and notification. (2 marks)





b. Explain how a Choice state can route invalid files to a failure path and valid files to processing. (1 mark)

A Choice state evaluates the validation result returned by the validation task.

If isValid is true, the workflow proceeds to the data transformation state.

Otherwise, the workflow transitions to a Fail state, stopping further processing of invalid files.

c. Explain how Retry and Catch can handle transient failures and permanent errors. (2 marks)

The purpose is to make your workflow handle errors automatically instead of stopping immediately.

d. Explain how the state machine can invoke Lambda functions or other supported AWS services. (1 mark)

Our goal is to connect the five processing states to actual AWS services.

AWS Step Functions uses Task states to invoke AWS Lambda functions and other AWS services.

- ValidateFile: Checks whether the CSV file is valid.
- TransformData: Cleans and transforms the data.
- CalculateTotalSales: Calculates total sales by product.
- StoreResults: Saves processed results to Amazon S3.
- SendNotification: Uses Amazon SNS to send a completion notification.

e. Explain how execution history helps troubleshoot a failed workflow. (1 mark) Expected Output: Draw the workflow or submit an Amazon States Language (ASL) definition

AWS Step Functions maintains an execution history that records every step performed by a state machine.

- Execution status: Shows whether the workflow succeeded, failed or is running.
- State details: Shows which task executed successfully or failed.
- Input and output: Displays the data received and returned by each state.
- Error messages: Helps identify why a task failed.
- Retry details: Shows repeated attempts after temporary failures.

Question 14: Step Functions Implementation (8 Marks)

A data engineering team has developed three processing functions:

• ValidateFile

• TransformData

• LoadResults The functions must execute in sequence.

Requirements:

• If validation fails, the workflow must stop.

• If transformation fails temporarily, the workflow must retry.

• If loading fails permanently, the workflow must record the failure.

Tasks:

a. Write an ASL JSON definition containing the three sequential tasks. Use placeholders for resource ARNs where necessary. (3 marks)

```
{
 "Comment": "Sales File Processing Workflow",
 "StartAt": "ValidateFile",
 "States": {
  "ValidateFile": {
   "Type": "Task",
   "Resource": "arn:aws:states:::lambda:invoke",
   "Parameters": {
    "FunctionName": "VALIDATE_LAMBDA_ARN",
    "Payload.$": "$"
   },
   "OutputPath": "$.Payload",
   "Next": "TransformData"
  },
  "TransformData": {
   "Type": "Task",
   "Resource": "arn:aws:states:::lambda:invoke",
   "Parameters": {
    "FunctionName": "TRANSFORM_LAMBDA_ARN",
    "Payload.$": "$"
   },
   "OutputPath": "$.Payload",
   "Next": "LoadResults"
  },
  "LoadResults": {
   "Type": "Task",
   "Resource": "arn:aws:states:::lambda:invoke",
```

```
    "Parameters": {
      "FunctionName": "LOAD_LAMBDA_ARN",
      "Payload.$": "$"
    },
    "OutputPath": "$.Payload",
    "End": true
  }
 }
}
```

b. Configure a retry policy for the transformation task with an appropriate error category, retry interval, and maximum attempts. (2 marks)

Step 1: Fix the errors

1. Open AWS Console → Step Functions.
2. Open your workflow in Workflow Studio.
3. Select each Lambda state and configure its Function name.
4. Select SendNotification and configure the SNS topic ARN.
5. Check that the red error symbols disappear.

Step 2: Create the state machine

1. Click Create.
2. Review the IAM execution role.
3. Confirm creation.

Step 3: Execute the workflow

1. Open Step Functions → State machines.
2. Select SalesProcessingWorkflow.
3. Click Start execution.
4. Enter the test input
5. Click Start execution again.

c. Configure a Catch transition for the loading task that routes to an error-handling state. (1 mark)

- Open AWS Step Functions → State machines.
- Select your state machine.
- Click Edit.
- In Workflow Studio, select the LoadResults state.
- Open Error handling → Catch.
- Click Add catcher.
- Enter the following values
- From the Flow section, add a Fail state.
- Rename it LoadFailed.
- Set Error to LoadFailed and Cause to Unable to load results.
- the Catch transition to LoadFailed.
- Click Save.

d. Explain how data from one task is passed to the next task. (1 mark)

- Open AWS Step Functions → State machines.
- Select your state machine and click Edit.
- Open Workflow Studio.
- Select the ValidateFile state.
- Under Input/Output, set:
- InputPath: $
- OutputPath: $.Payload
- Select the TransformData state.
- Set:
- InputPath: $
- OutputPath: $.Payload
- Select the LoadResults state.
- Set:
- InputPath: $
- OutputPath: $.Payload

- Click Save.

e. Explain how you would test successful and failed executions. (1 mark)

- Open AWS Console → Step Functions → State machines.
- Select your state machine.
- Click Start execution.
- Enter valid input
- Click Start execution and check the status.
- Start another execution with invalid input
- Open Execution history to check the result and errors.
-

SECTION F: MICROSOFT AZURE || Maximum Marks: 15

Question 15: Azure Data Lake Storage and Security (5 Marks)

A company needs to store confidential business data in Azure for analytics. Perform or document the following:

a. Configure or describe a storage account with hierarchical namespace enabled for Azure Data Lake Storage Gen2. (1 mark)

- In Azure Portal, search for Storage accounts and select Create.
- Choose Subscription and Resource group (e.g. rg-cloud-data-lab).
- Enter a globally unique lowercase storage account name (for example, adlscloudlab plus unique digits).
- Choose an allowed Region, Standard performance and a suitable redundancy option such as LRS for a lab.
- Open Advanced and enable Hierarchical namespace (Data Lake Storage Gen2).
- Under Networking, choose network access appropriate to your lab; for confidential production data, prefer private endpoints and restricted networks.
- Keep public blob anonymous access disabled; review the remaining settings and click Review + create, then Create.
- Open the deployed storage account and verify hierarchical namespace in Settings > Configuration (or the account properties).

b. Create a filesystem or container and upload a sample dataset. (1 mark)

- Open the new storage account > Data storage > Containers.

- Click + Container. Enter name customer-data (lowercase) and keep Anonymous access level = Private. Click Create.
- Open customer-data. Click Upload and select a local CSV named customer_transactions.csv.
- Use the sample content below if you need a safe demonstration dataset.
- Click Upload and verify the blob appears in the container.

c. Explain how Microsoft Entra ID and Azure role-based access control can manage access to the data. (2 marks)

- In Azure Portal, search for Microsoft Entra ID > Users or Groups. Select an existing authorized test user/group, or create a lab group if you have permission.
- Open Storage accounts > your ADLS Gen2 account > Access control (IAM).
- Click Add > Add role assignment.
- Under Job function roles, select Storage Blob Data Reader for read-only data access, or Storage Blob Data Contributor only for users who must upload/modify files.
- On Members, select User, group, or service principal; choose the authorized test identity.
- Review + assign. Wait briefly for role assignment propagation.
- For tighter scope, open the customer-data container > Access control (IAM) if supported, and assign the role at container scope rather than the entire account.
- Use Entra-authenticated access when browsing data; remember Azure management roles such as Reader do not automatically grant blob data read access.

d. Explain how you would verify that unauthorized users cannot access the dataset. (1 mark)

- Choose a separate authorized test account that has no blob data role assignment and no applicable ACL permissions. Do not use a real person's account without permission.
- Open a private/incognito browser window and sign in as that test account.
- Try to open customer-data/customer_transactions.csv using the Azure Portal data browser or a suitable Entra-authenticated tool.
- Confirm that the data access is denied (for example, an authorization error); the account may still be visible if the user has management-plane Reader access.
- Sign in as the permitted test user and verify that reading the same file succeeds**.**

Question 16: Azure VM Troubleshooting (5 Marks)

A virtual machine is running in Azure, but administrators cannot connect to it. Perform or document the following:

a. Verify the VM's provisioning and power states. (1 mark)

- In Azure Portal, search for Virtual machines and open the affected VM.
- On Overview, check Status/Power state (for example, Running or Stopped) and Provisioning state (for example, Succeeded).
- Open Help > Resource health to check whether Azure reports a platform problem.
- Open Monitoring > Activity log to inspect recent VM updates or failed operations.
- If stopped or deallocated, start the VM only if you are authorized and understand the cost implications.

b. Explain how Network Security Group rules can prevent connectivity. (1 mark)

- Open the VM > Networking or Network settings. Identify the attached network interface and any subnet NSG.
- Review inbound security rules and effective security rules for the NIC.
- For Linux SSH check TCP 22; for Windows RDP check TCP 3389.
- Verify that a higher-priority Deny rule is not overriding the required Allow rule. Lower priority numbers are evaluated first.
- If a direct inbound rule is required for a controlled test, restrict Source to a trusted public IP/CIDR; do not use Any/0.0.0.0/0 for management ports.
- Also check the VM guest firewall and whether the SSH/RDP service is running**.**

c. Explain how subnet and routing configuration affect connectivity. (1 mark)

- Open the VM > Networking > Network interface and note its Virtual network, Subnet, private IP and public IP (if any).
- Open Virtual networks > selected VNet > Subnets and inspect the subnet configuration.
- Open the VM NIC > Effective routes (or Network Watcher > Next hop) and inspect routing to/from the client network.
- If using direct internet access, verify that a reachable public endpoint and a valid return path exist; a private-only VM cannot be reached directly from the public internet.

- Check whether a route table/UDR sends traffic through a firewall or network virtual appliance that might block the connection.

d. Recommend a secure troubleshooting method that avoids unnecessarily exposing the VM to the public internet. (2 marks)

- Prefer Azure Bastion, VPN, ExpressRoute or another approved private access path instead of assigning a public IP and opening management ports to the internet.
- For Azure Bastion: open the VM > Connect > Bastion. If Bastion is not already deployed, deployment may require a dedicated AzureBastionSubnet, suitable IP resources, permissions and additional charges.
- Select the approved authentication method (for example SSH private key for Linux or approved credentials) and connect.
- If the connection still fails, use Network Watcher > Connection troubleshoot / IP flow verify and review NSG rules, routes and guest service status.
- Use VM > Help > Run command (where permitted) to inspect services without direct inbound SSH/RDP; capture diagnostics, then restore any temporary troubleshooting changes.

Question 17: Azure Data Integration (5 Marks)

A company receives a CSV file containing daily customer transactions and needs to prepare it for analysis.

Design a solution using Azure services.

a. Select an appropriate service to orchestrate data movement. (1 mark)

- In Azure Portal, search for Data factories > Create.
- Select subscription, resource group, a supported region and a unique data factory name (for example, adf-cloud-data-lab).
- Review + create and deploy the factory.
- Open the factory and click Launch Studio.
- Select Author > Pipelines > + New pipeline and name it pl_daily_customer_ingestion.

b. Explain how to copy the CSV file into Azure Data Lake Storage Gen2. (1 mark)

- Ensure your sample CSV is available at an accessible source, such as an input container/folder in Azure Blob Storage or ADLS Gen2.

- In ADF Studio > Manage > Linked services, create a linked service to the source storage and one to the destination ADLS Gen2 account.
- Use Managed identity authentication when supported; assign the Data Factory managed identity the required Storage Blob Data Reader role on the source and Storage Blob Data Contributor role on the destination.
- In Author, create a source dataset using DelimitedText and select the source linked service, container and file. Mark First row as header if appropriate.
- Create a sink dataset using Azure Data Lake Storage Gen2 / DelimitedText, selecting the destination linked service and target filesystem/path.
- Drag Copy data onto the pipeline. In Source, choose the CSV dataset. In Sink, choose the destination dataset. Configure column mapping if needed.
- Click Validate, then Debug for a test run. Confirm success in Output and inspect the destination container for the copied file.

c. Describe how to organize files by ingestion date. (1 mark)

- . In the ADLS Gen2 destination, use a directory structure such as raw/customer_transactions/ingestion_date=2026-10-28/.
- For a repeatable pipeline, parameterize the sink folder path using a pipeline parameter or ADF expression.
- Example dynamic folder expression: @concat('raw/customer_transactions/ingestion_date=',formatDateTime(utcNow(),'yyyy-MM-dd'))
- In the sink dataset or Copy activity dataset parameters, pass the generated folder path; verify that the output is written under the expected date folder.

d. Explain how to handle malformed records and missing values. (1 mark)

- Inspect the CSV schema and identify required columns such as TransactionID, CustomerID, TransactionDate and Amount.

- For simple type-conversion problems in Copy data, review the activity's Fault tolerance / skip incompatible rows options where available and capture rejected-row logs if configured.
- For business validation, add a Mapping Data Flow (or Databricks transformation) after ingestion.
- Use derived-column rules to trim text, convert dates/amounts and handle permitted defaults; use conditional split to separate valid and invalid records.
- Write valid rows to a curated folder and rejected rows to a quarantine folder with an error reason.
- Do not silently replace missing required identifiers; quarantine those records for review.

e. Explain how to monitor pipeline execution and validate data completeness. (1 mark)

- In ADF Studio, select Monitor > Pipeline runs and locate the latest run of pl_daily_customer_ingestion.
- Check status, start/end times and duration; open Activity runs to inspect Copy data metrics such as rows read, rows copied and errors.
- Compare source record count with successful target records plus rejected records; confirm required files and expected date folders exist.
- Investigate differences and failures using activity output and configured diagnostic logs.
- For recurring loads, add a schedule trigger only after a successful manual test and configure alerting through Azure Monitor where appropriate.